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Palaeomagnetism of igneous rocks of the Lake District (Caledonian) terrane, northern England: Palaeozoic motions and deformation at a leading edge of Avalonia

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The Lake District terrane of northern England comprises Upper Cambrian–Silurian sediments and volcanics accumulated at the northern margin of the Avalonian Plate during growth and demise of the Iapetus Ocean. Ocean closure and suturing resulted in Late Ordovician and Acadian tectonism and were accompanied by emplacement of a large regional batholith. Palaeomagnetic study of intrusive igneous rocks, including application of thermal demagnetization, field tests and principal component analysis, identifies a history of Ordovician to Devonian magnetization. Late plutons (Shap and Skiddaw granites and/or aureoles) record a shallow dipolar (A3) axis (mean declination/inclination ($D/I = 278/+17^\circ$) dating from emplacement in late Early Devonian times (*c.* 395 Ma). Although this axis is recorded as a sporadic overprint in older rocks, no pervasive remagnetization is attributable to batholith emplacement. Instead, the Carrock Fell Complex Layered Gabbros have a mid- to late Ordovician (A1) remanence ($D/I = 17{\cdot}4/-58{\cdot}1^\circ$, 36 samples, $\alpha_{95} = 4{\cdot}8^\circ$) predating regional F2 folding. Later events in this igneous complex comprise the Carrock Fell Granophyre with a post-folding Ordovician remanence, and Round Knott Dolerite with a remanence linked to hydrothermal alteration late in the Ordovician magmatic episode. A Late Ordovician (Ashgill) palaeofield is also defined by remanence (A2) in the Threlkeld–St John's Microgranite and aureole (438 Ma, $D/I = 236{\cdot}5/63{\cdot}3^\circ$, 41 samples, $\alpha_{95} = 4{\cdot}7^\circ$). Other intrusions carrying a remanence predating the Acadian deformation include the Great Cockup Picrite (458 Ma, $D/I = 43{\cdot}2/-31{\cdot}8^\circ$, 31 samples, $\alpha_{95} = 7{\cdot}7^\circ$) and basic intrusives in the aureole of the Eskdale Granite (429 Ma, $D/I = 174{\cdot}5/25{\cdot}8^\circ$, 32 samples, $\alpha_{95} = 8{\cdot}8^\circ$).

Collectively the palaeomagnetic data from this terrane identify a hairpin in the apparent polar wander path during Late Ordovician (Caradoc–Ashgill) times corresponding to 'soft' closure of the Iapetus suture and accompanying deformation. The same motion is recognized in contemporaneous data from the Welsh Caledonides where declinations are rotated by *c.* 55° relative to contemporaneous results from the Lake District. Adjustment for this (probable late Acadian) rotation beings fold trends of the Paratectonic Caledonides into alignment and identifies a parallel mid- to late Ordovician destructive plate margin comprising forearc (Lake District) and backarc (North Wales). This arc was oriented latitudinally in mid-southerly latitudes during formation and the bulk of the magmatism occurred during a single normal-polarity chron. The relationships between magnetization and folding in both the Lake District and Welsh Borderlands identify the importance of Late Ordovician deformation along this arc during collision of Avalonia and Laurentia. Arc-related volcanism was succeeded in Silurian times by parallel foreland basins embracing the Welsh Basin and southern Lake District as the Laurentian Plate overrode the Avalonian Plate. © 1997 John Wiley & Sons, Ltd.

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1. INTRODUCTION

Assembly of the supercontinent of Pangaea in Palaeozoic times involved collisional docking of large continental blocks with a long Precambrian history (Gondwana, Laurentia, Baltica and Siberia) plus a number of smaller continental blocks, mostly with more limited Late Precambrian histories, such as those now forming the European Massifs. The latter include the Western Avalonian Plate (now sited at the eastern margin of North America), the Eastern Avalonian Plate including the basement of southern Britain, the

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Armorican Massif of Brittany and the Bohemian Massif of central Europe. The former continuity between these terranes is a subject of topical debate but faunal evidence has tended to assign them to a peri-Gondwanic location in early Palaeozoic times (Cocks and Fortey, 1990; Scotese and McKerrow, 1990) broadly compatible with high southerly palaeolatitudes in early Ordovician times (Van der Voo, 1993).

The Early Palaeozoic motions between Eastern Avalonia and Laurentia document progressive closure of the Iapetus Ocean between these blocks (Scotese and McKerrow, 1990), culminating in a 'soft' collision by early Silurian times and final welding during the Acadian Episode of the Caledonian Orogeny in Late Silurian–Early Devonian times (Cocks and Fortey, 1990). Volcanic products of Ordovician ocean closure emplaced above a southerly dipping subduction zone (Fitton and Hughes, 1970) are well represented in southern Britain and have been the focus of recent palaeomagnetic studies (see summaries in Trench and Torsvik (1991) and Channell *et al*. (1992a)). The results form the present basis for correlating Eastern Avalonia with the larger blocks and document the history of ocean closure and progressive collisionsal docking that led to formation of Pangaea.

The two major outcrops of Eastern Avalonia comprise the Lake District of northern England and most of Wales (referred to hereafter as the Welsh Basin). Although palaeomagnetic results from these two terranes have hitherto been considered as a single entity (Trench and Torsvik, 1991), the regions between (and most of the remainder of the Eastern Avalonian Block) are widely covered by post-Caledonian sedimentary successions; hence their tectonic continuity in pre-Acadian times requires evaluation. Furthermore, there are major gaps in the present data especially during the critical interval of mid-to-late Ordovician times when faunal evidence suggests that ocean closure was complete and subduction had effectively ceased.

These problems can be addressed by study of the intrusive igneous rocks spanning these times in both regions. A study of such rocks in the Lake District inlier is the subject of this paper. Although the age and tectonic constraints on such bodies are not as precise as controls on the stratified rocks, which will therefore more often form the anchor points in any apparent polar wander (APW) path, it is found here that data from the intrusions widely support the key data; they also serve to document APW during intervals unrepresented by suitable stratified formations. In Section 5 the data are compared with contemporaneous results from the Welsh Basin (Channell *et al*., 1992a; Piper, 1995) to document kinematic history of the Avalonian Plate during closure of the Iapetus Ocean and concomitant Caledonian tectonism.

2. GEOLOGICAL FRAMEWORK

The Lake District terrane is the northernmost outcrop linked to the Avalonian Plate in Britain. It is located immediately south of the Solway Line (Figure 1B) along which Eastern Avalonia merged with Laurentia. Immediately to the north, a thick prism of Lower Palaeozoic sediments with ophiolitic components was accreted during a complex history of oblique impingement onto the Laurentian margin and now forms the Southern Uplands terrane (Bluck, 1985).

During Late Cambrian and Early Ordovician times this margin of Eastern Avalonia was a site of marine sedimentation. The oldest exposed rocks are the Skiddaw Group comprising >5000 m of graptolitic mudstones, greywackes and sandstones. Deposition ceased during the D. *bifidus* Biozone of Upper Llanvirn times (Downie and Soper, 1972; Wadge, 1978). A subsequent volcanic episode in Middle to Late Ordovician times produced up to 2500 m of basalt and andesite lavas which now form a northern outcrop of Ordovician volcanics in the Lake District (Eastwood *et al*., 1968; Figure 1) assigned to the Eycott Volcanic Group (EVG). Emplacement of the Carrock Fell Layered Gabbro Complex and a variety of smaller intrusions included in this study (Figure 1 and Table 1) was broadly contemporaneous with this event.

At about the same time, *c*. 7000 m of volcanics constituting the Borrowdale Volcanic Group (BVG) were extruded in the central Lake District during a volcanic episode incorporating piecemeal collapse of calderas (Branney and Soper, 1988), and completed by Early to Middle Caradoc times (D. *multidens* Biozone (Wadge, 1978)). This stratigraphic age is supported by a 457 ± 4 Ma Sm–Nd mineral age (Thirlwall and

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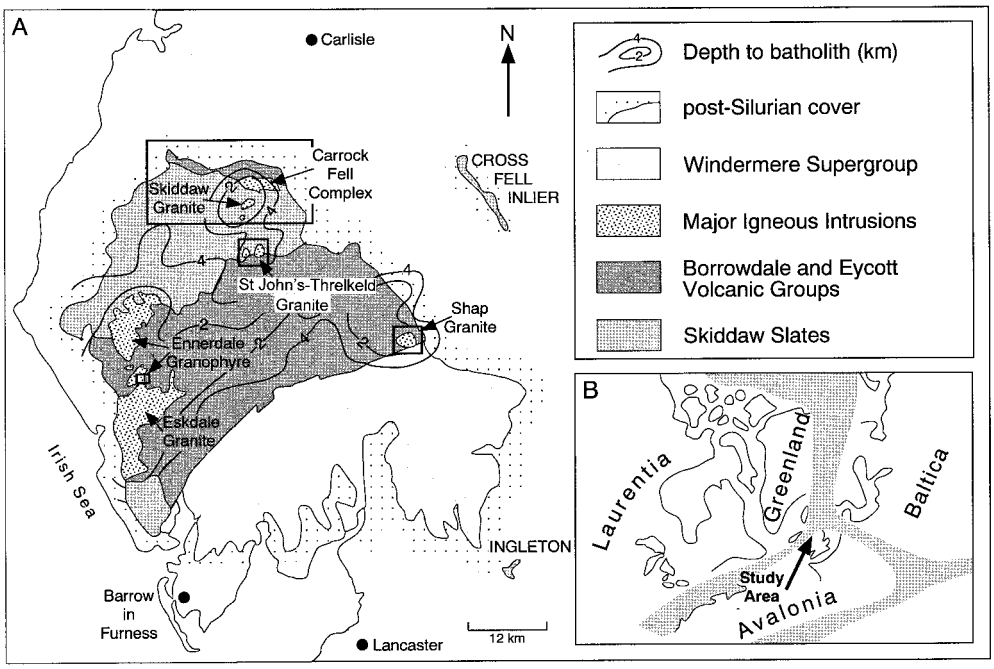


Figure 1. (A) The Lower Palaeozoic Lake District inlier, northern England, showing major rock divisions and intrusions sampled in this study. The depth to the regional batholith is derived from the Bouguer gravity anomaly after Lee (1984). The boxed areas appear as expanded maps with sampling sites indicated in subsequent figures. (B) Cartoon figure showing the situation of the Lake District inlier within the Avalonian Plate in the reconstruction of the North Atlantic continents attained following closure of the Iapetus Ocean by Devonian times

Table 1. Summary of the Palaeozoic stratigraphic succession and deformation events, Lake District terrane, northern England

Major rock divisions	Lithologies	Environment
Carboniferous Limestone (Lower Carboniferous, Courceyan)		Marine shelf
	Unconformity, F3 (Acadian) Episode	
Windermere Supergroup (4000 m, Ashgill–U. Silurian)	Turbidies succeeding a shelf sequence	Unstable marine basin formed following 'soft' collision of Avalonia and Laurentia
	Unconformity, F2 (U. Ordovician) Episode	
Borrowdale Volcanic Group (up to 7000 m, mid-Ordovician) Eycott Volcanic Group (up to 2500 m, mid-Ordovician)	Tholeiitic and calc-alkaline volcanics	Magmatism above SE subducting Iapetus Ocean crust
	Unconformity, F1 Episode	
Skiddaw Slate Group (> 5000 m, base not visible). Cambrian–mid-Ordovician	Greywackes and graptolitic mudstones	Continental margin of E. Avalonia

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Fitton, 1983). The BVG and EVG events (Table 1) have a chemistry compatible with a relatively evolved continental margin environment (Fitton and Hughes, 1970; Millward *et al*., 1978). Together with associated intrusives, they record the final phase of subduction-related magmatism at this margin of Avalonia.

Following cessation of volcanism, a transgressing sea deposited limestones, shales, mudstones and greywackes between Late Caradoc and Ludlow times. This succession forms the Winderemere Supergroup and makes up the Silurian outcrop in Figure 1. During burial and subsequent Acadian deformation the Lower Palaeozoic rocks of the Lake District were metamorphosed to no higher than lower greenschist grade, in common with the paratectonic Caledonides of Wales; this was insufficient to comprehensively remagnetize pre-Acadian rocks which are reported widely to retain an Ordovician remanence record (Briden and Morris, 1973; Piper *et al*., 1978; Piper, 1979; Channell and McCabe, 1992).

In addition to the basic plutonism noted above, acidic magmatism occurred at various stages between the intense mid-Ordovician volcanic events and the culmination of tectonic activity in the Acadian episode. A general distinction is recognized between earlier Na-rich activity and later more K-rich activity (Piper *et al*., 1978). The Carrock Fell and Ennerdale Granophyres are typical sub-volcanic plutons, whilst the Threlkeld–St John's body is a microgranite with small phenocrysts of plagioclase and pyroxene suggesting near-liquidus compositions (O'Brien *et al*., 1985). The Eskdale, Skiddaw and Shap plutons are coarser-grained bodies incorporating at least two stages of crystallization and more abundant K-feldspar and quartz.

The regional Bouguer gravity anomaly highlights a large granite batholith underlying much of the Lake District (Bott, 1974; Lee, 1984), which comes to the surface as the Eskdale Granite in the western Lake District (Figure 1). Elsewhere it lies at shallow depths with additional cupolas recorded by late plutons (Shap and Skiddaw) where only roof zones are exposed (Figure 1A). Whilst the gravity expression is comparable to that of the buried Weardale (410 ± 10 Ma) and Wensleydale (400 ± 10 Ma) granites beneath the Pennines of central Britain and suggests an Early Devonian origin, the Lake District Batholith has a complex history at outcrop (Firman, 1978) and emplacement probably spanned the interval between mid-Ordovician and Early Devonian times: most or all of the granophyres are genetically linked to the volcanic episode whereas the Shap and Skiddaw Granites are of late Early Devonian age and post-date all but the last and most brittle deformation (Shepherd *et al*., 1976; Wadge *et al*., 1978).

3. FIELD AND LABORATORY METHODS

3a. Strategy of investigation and previous studies

The granitic batholith represents an extensive late-Caledonian thermal influence with effects widely recorded by metasomatism in the acid plutons and by hydrothermal mineral veins in the country rocks (O'Brien *et al*., 1985). Hence palaeomagnetic analysis needs to address the extent of magnetic overprinting by this event. Unfortunately the granites are mostly composed of coarse leucocratic quartz–feldspar assemblages unsuitable for palaeomagnetic study. Accordingly they are investigated here (Skiddaw, Shap and Eskdale Granites) largely in terms of thermal overprinting of adjoining country rocks; in addition the Shap Granite is reddened and partially retains a stable ancient remanence whilst the Ennerdale Granophyre is intruded by basic bodies probably only a little younger than the age of granophyre emplacement. Collective investigation of these plutons allows the partial overprinting effects of the Late Caledonian thermal event to be recognized in the older Ordovician igneous bodies. With the exception of gentle folding linked to the Variscan Episode and post-Variscan rifting, the region has been quiescent since Caledonian times, although uplifted with removal of 2–3 km of overburden since early Tertiary times.

The Carrock Fell Complex was the subject of earlier study employing a.f. demagnetization and endpoint analysis (Briden and Morris, 1973; Morris, 1973). Since the complex was then considered to be a post-tectonic event, the data were evaluated *in situ*; more recent evidence shows that the layered gabbros predate F2 folding (Roberts, 1971; Harris and Dagger, 1987). The present study includes resampling and stepwise thermal demagnetization of the Carrock Fell Gabbros and contacts, with fabric also investigated using anisotropy of

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magnetic susceptibility (AMS), and study of the later stages of magmatism (Carrock Fell Granophyre, Round Knott Dolerite and cross-cutting dyke swarm). Wider expressions of the mid-Ordovician magmatism to the west, including the Great Cockup Picrite, are also included in this investigation.

Palaeomagnetic study of the Lake District inlier by Morris (1973) and Briden and Morris (1973) on the EVG and Carrock Fell Complex was extended to the BVG outcrop of the central Lake District by Faller *et al.* (1977) and Piper (1979). More recent investigations employing refined demagnetization and analysis accompanied by palaeomagnetic field tests have focused on the BVG and recognize variable degrees of partial overprinting (Channell and McCabe, 1992; Piper *et al.*, 1997).

No systematic study has previously been undertaken of the widespread intrusive igneous bodies, although results from a number of isolated studies are given by Morris (1973) and Piper *et al.* (1978). They have not been considered as priorities for palaeomagnetic study owing to age and orientation uncertainties. These latter are, however, mostly small and the intrusions are therefore targets for investigation because suitable stratigraphic units are so few (and essentially limited to the EVG and BVG). The intrusions tend to be magnetically less stable than equivalent volcanics (Morris, 1973; Piper *et al.*, 1978, the Carrock Fell Layered Gabbros are a notable exception) probably because they are more commonly dominated by larger magnetite grain sizes and multi domains. Higher failure rates therefore accompany these studies and some samples have a remanence dominated by Recent field components. Nevertheless, numerous intrusive igneous samples show a coherent remanence structure unobscured by Variscan or later overprinting and it is with these components that the present study is primarily concerned.

The palaeomagnetic samples were collected in the field typically over several metres of outcrop (located in the data tables by their National Grid reference numbers) using portable drills. Cores were oriented *in situ* using Sun and magnetic compasses and cut into 2·4 cm length cylinders for palaeomagnetic processing.

3b. Laboratory methods

Natural remanent magnetizations (NRMs) were measured with 'Minispin' or SQUID magnetometers and thermally demagnetized in steps of 50 or 100°C to 500°C; subsequent treatment continued in steps of 15 or 20°C to 580°C (or 680°C if haematite carriers were evident). The whole procedure is conducted within sets of large Rubens coils in which the ambient magnetic field is continuously nullified to minimize secondary remanence acquisition. Components of magnetization comprising the NRM were recognized by study of orthogonal projections derived from results of the progressive thermal treatment. Characteristic magnetizations (ChRMs) are isolated and directions of components defined by three or more successive steps of demagnetization determined using principal component analysis.

Ancillary rock magnetic investigations included study of anisotropy of magnetic susceptibility (AMS) conducted on selected samples from the Skiddaw Slates, Carrock Fell Layered Gabbros and the Round Knott Dolerite, and thermomagnetic determinations (saturation remanence, J_s, versus temperature determined with an automated Curie balance) on representative samples of each facies to help identify magnetic mineral constituents.

4. PALAEOMAGNETIC RESULTS

4a. The Shap Granite

This 'granite' is a pink porphyritic adamellite emplaced at the eastern margin of the Lake District inlier (Figure 1) into Borrowdale Volcanics and Silurian mudstones; the latter were deformed during the Acadian episode prior to granite intrusion. The pluton is a steep-sided body with a 5 km^2 outcrop (Locke and Brown, 1978). Emplacement is well constrained in stratigraphic terms: the youngest invaded country rocks are deformed sediments of Ludlow age; pebbles of granite found in nearby Lower Carboniferous sediments identify uplift and unroofing during Devonian times following the Acadian episode.

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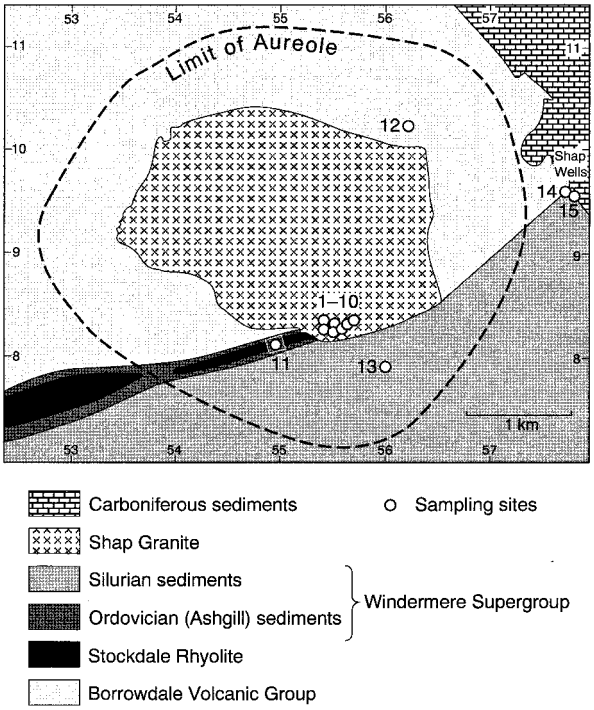


Figure 2. Geological map of the Shap Granite and aureole, northern England, simplified after Wadge *et al.* (1978) and showing locations of the palaeomagnetic sampling sites of this study. The numbers of National Grid lines are shown at the margins

Biotite in the contact aureole has grown along a regional Acadian cleavage but predates a late sub-horizontal crenulation cleavage (Boulter and Soper, 1973). The structural evidence thus defines an emplacement which is essentially post-tectonic to the Acadian Episode and has an older limit, of early Gedinnian age (early Early Devonian (Wadge *et al.*, 1978)). The granite is dated 397 ± 7 Ma (mean of K–Ar determinations on biotite) and 394 ± 3 Ma (Rb–Sr isochron (Wadge *et al.*, 1978)); these ages are confirmed by U–Pb data from zircons yielding an age of 390 ± 7 Ma (Pidgeon and Aftalion, 1978), indicating emplacement shortly before the Early–Middle Devonian boundary at *c.* 386 Ma (Harland *et al.*, 1990).

Palaeomagnetic studies on 207 cores from 10 sites distributed at large fresh quarry faces in the pluton (Figure 2) have been conducted. Approximately equal numbers of cores were drilled from adamellite intrusion and xenoliths. The xenoliths are BVG lavas and Silurian mudstones variably digested and recrystallized within the igneous host. Reflecting their diverse origins, the xenoliths have a wider range of intensities of magnetization (4×10^6–6×10^3 A m^2/kg) than the adamellite (7×10^6–8×10^4 A m^2/kg) but are typically more strongly magnetized by an order of magnitude. Thermomagnetic study identifies discrete magnetite Curie points linked to accessory grains of this mineral. This behaviour is also typical of xenoliths and the saturating phase undergoes little alteration on heating in air (Figure 3). There is no appreciable indication of the red haematite pigment in the orthoclase phenocrysts; since the thermomagnetic curves are dominated by the magnetite this is not unexpected because the saturation remanence of haematite is only *c.* 2% of that of magnetite.

Some samples at all locations have dominant steep positive ('B') components resident in the magnetite spectrum (sample 3-13 in Figure 4) which generally do not quite converge on the origin; a small residual component is evident but not resolved. This higher blocking temperature (h.b.t.) component is recognized in other samples with little or no B component where it is present as a weaker magnetization of lower stability with easterly or westerly declination and shallow inclination (samples 1–4 and 8–12 in Figure 4). Thus both

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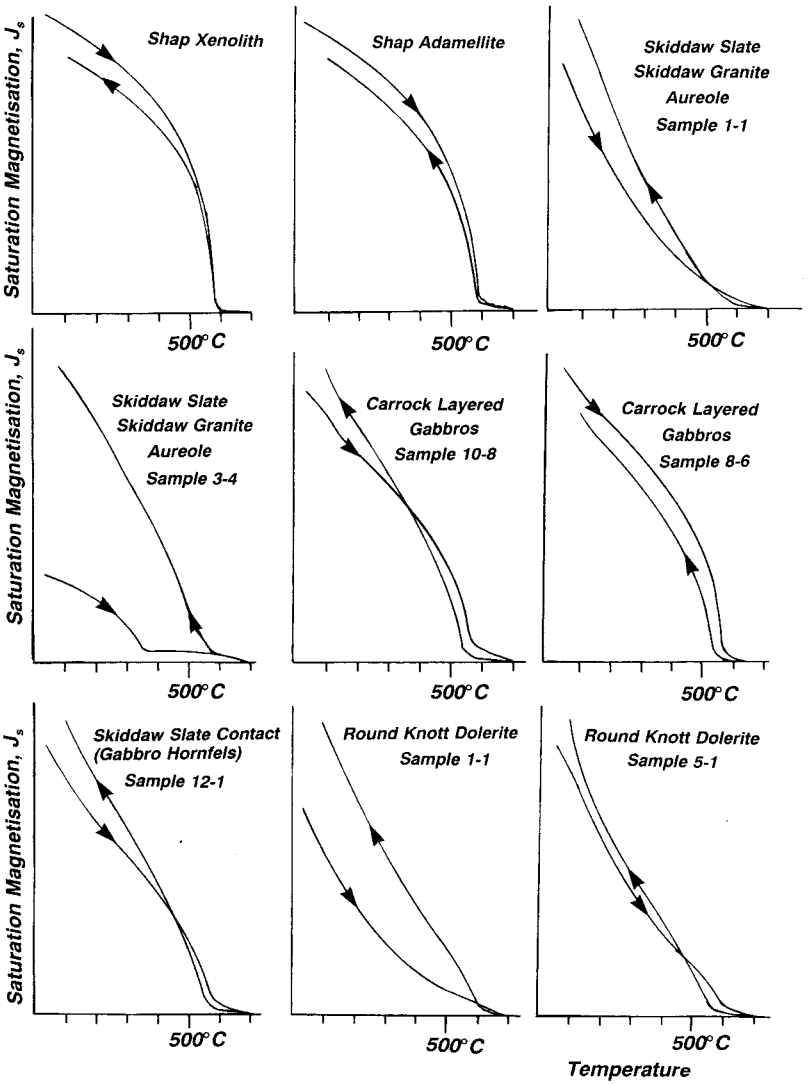


Figure 3. Thermomagnetic determinations (saturation remanence, J_s, against temperature in °C) from some rock types of this study

polarities of this ('A3') magnetization are recorded. In a few cases (e.g. sample 7-11 in Figure 4) each polarity appears to be subtracted successively; cooling through the magnetite blocking temperature spectrum has probably recorded polarity reversal in these examples. Definition of the A3 component deteriorates as the relative magnitude of the B component increases; when B dominates the spectrum, A3 is often still recognizable but vector definition is erratic. B is unblocked at a wide range of temperatures between 100 and 500°C, whilst A3 is unblocked at temperatures ranging from 400 to 630°C, implying some presence in the pigmentary haematite.

The distribution of component directions shows slight smearing between the B and westerly A3 groups (Figure 4) suggesting that they have not always been fully resolved; however, mean directions of the two A3 polarities (calculated for inclination $I < 25°$) are nearly antiparallel. The 95% confidence cones overlap but the angle between the normal and reversed axes (11.5°) is slightly greater than that required for a positive reversal test (9·6° (McFadden and McElhinny, 1990)). The other feature of the palaeomagnetic record in this

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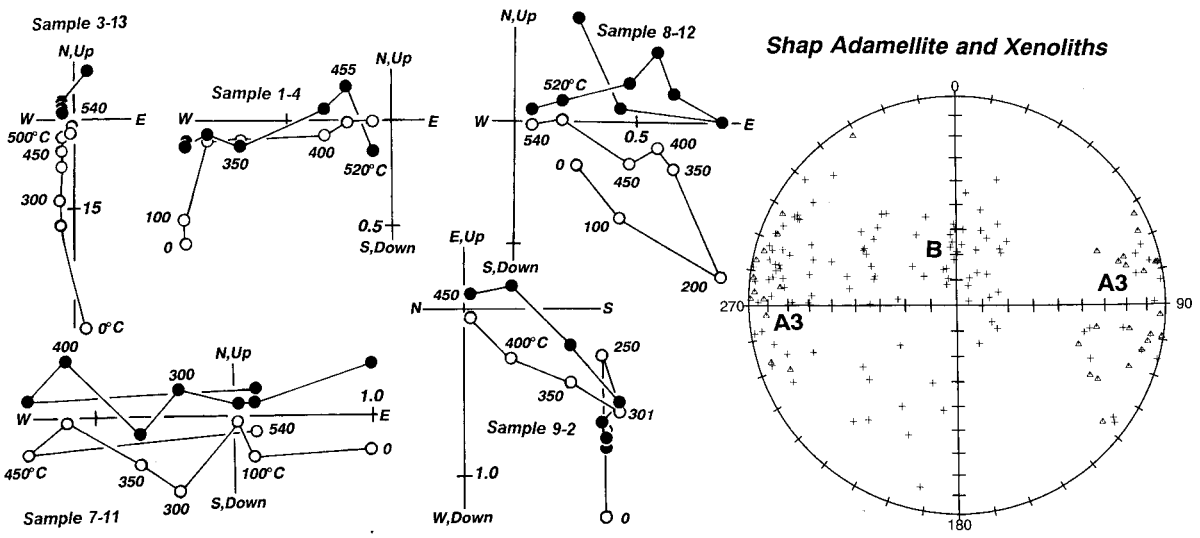


Figure 4. Typical results of progressive thermal demagnetization of samples from the Shap adamellite and xenoliths of country rock illustrated (*in situ*) as orthogonal projections. The closed circles are projections onto the horizontal plane and the open circles are projections onto the vertical plane; temperatures are in °C and intensities are $\times 10^{-5}$ A m^2/kg. The equal-area projection shows the distribution of all components of magnetization identified in samples of the adamellite and xenoliths; the northerly positive directions are the lower blocking temperature (B) magnetizations and the easterly and westerly shallow directions are the higher blocking temperature (A3) magnetizations. Lower hemisphere (positive inclination) directions are shown as crosses and upper hemisphere (negative inclination) directions as open triangles

pluton is sporadic preservation, mostly at site 9, of a poorly defined SW component unblocked by 400–500°C (sample 9-2 in Figure 4). This was recognized by earlier a.f. demagnetization of 12 samples from the same location (Piper *et al.*, 1978) but is evidently not the ChRM in the pluton.

The mean of the B components (declination/inclination $D/I = 345/72°$, 54 components, $\alpha_{95} = 5{\cdot}5°$) is close to the present magnetic field ($D/I = 355/71°$) and therefore interpreted to be of Recent viscous origin. The presence of both polarities in the (predominantly magnetite-resident) A3 magnetization suggests that it is the ChRM of primary cooling-related origin. This interpretation can potentially be tested in this case by two palaeomagnetic field tests, namely (i) a contact test on the country rocks of the aureole and (ii) a conglomerate test on overlying Lower Carboniferous conglomerates.

Basaltic lavas belonging to the BVG were sampled in the Shap Blue Quarry and adjoining stream sections (Figure 2). In these rocks metasomatic influence of the pluton is shown by development of epidote and other low-grade minerals. Magnetizations (site 12) are dominated by large lower blocking temperature (l.b.t.) components apparently acquired in the present field and comparable to the B component in the pluton (Figure 5). A residual convergent magnetite-resident component was recovered in 13 of 17 cores following subtraction of the B component, with a direction similar to the westerly shallow A3 remanence in the pluton (Figure 5 and Table 2). This is widely removed from the ChRM in the BVG which is directed NNE (Piper *et al.*, 1997) (the lavas have low dip at this location); the primary remanence in the lavas therefore appears to have been completely overprinted by thermal effects of the pluton.

The aureole also cuts the Stockdale Rhyolite, the last expression of Borrowdale volcanism in the Lake District; this has been sampled at site 11 (Figure 2). Magnetizations are very weak (*c.* 10^{-7} A m^2/kg) but progressive thermal demagnetization above 500°C again isolates similar components to the A3 magnetization recognized in the pluton (Figure 5 and Table 2); in this case examples of both polarities are recorded with no prominent record of the B magnetization. No vestiges of the primary (steep negative-directed) remanence found in the rhyolite elsewhere (Piper *et al.*, 1997) are observed; hence this unit has also been overprinted by thermal influence of the pluton.

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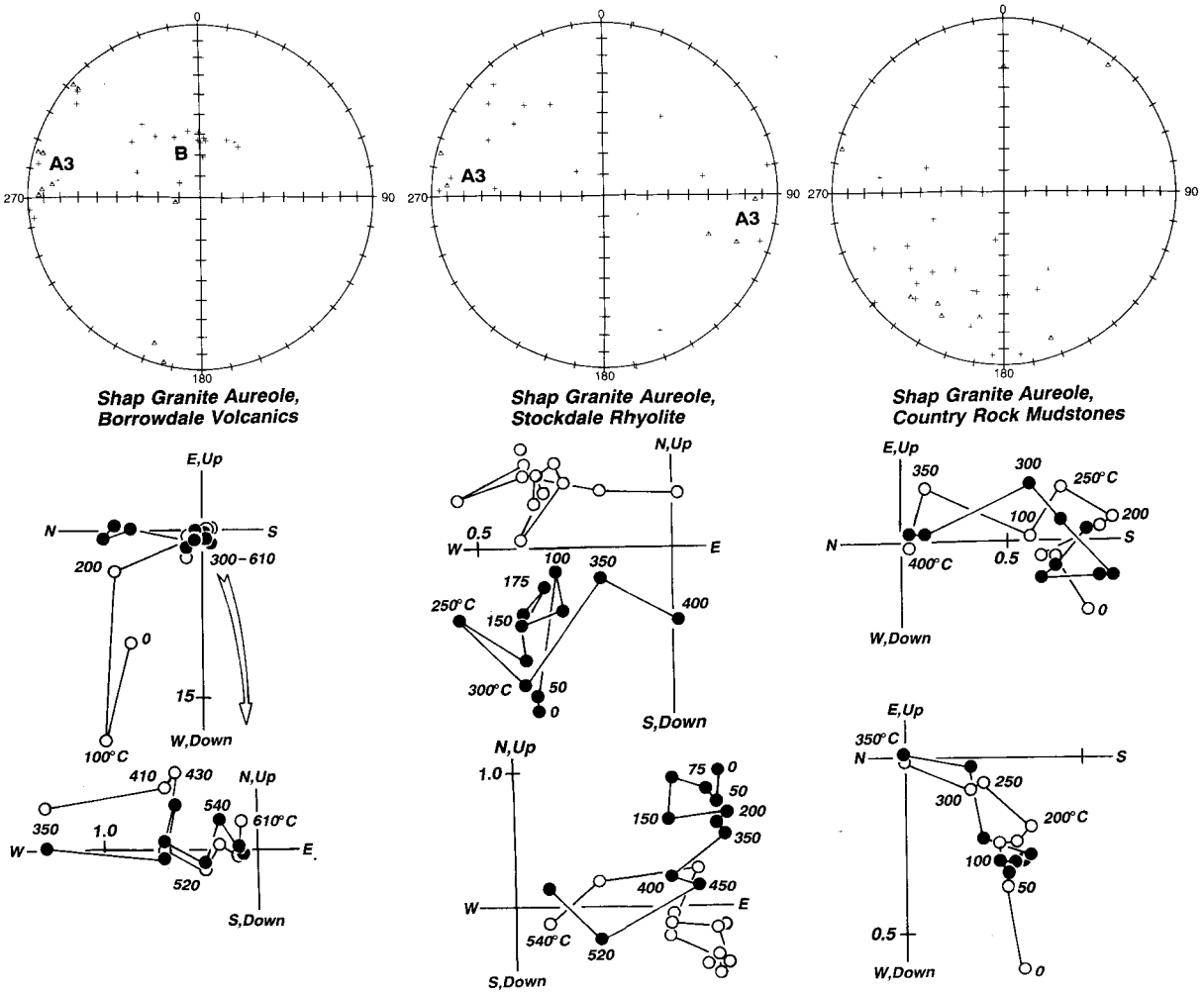


Figure 5. Equal-area projections showing directions (*in situ*) of components identified by progressive thermal demagnetization of the three rock facies within the Shap Granite aureole. Demagnetization results are shown by orthogonal projections of Borrowdale Volcanic Group Lava (low blocking temperature Recent field component excluded), the Stockdale Rhyolite and Silurian mudstones; symbols are the same as in Figure 4. Sample sites are shown in Figure 2

Silurian mudstones (site 13, Figure 2) at the southern margin of the pluton proved to be a less suitable facies for a contact test. Weak remanence is fully subtracted by 350–400°C and therefore probably resident in pyrrhotite (Figure 5). Most ChRMs are of reversed polarity but variable direction; they correspond broadly to the predicted field direction in Late Palaeozoic times and include the minor direction recognized in the pluton. Only two of 17 cores yield an h.b.t. component approximately corresponding to the A3 remanence in the pluton. Diagenetic mineral growth in the mudstones (possibly during the last stages of cleavage development in the Acadian Episode (see Boulter and Soper, 1973)) has evidently obscured significant record of overprinting by the igneous body.

Detritus from the Shap Granite occurs in basal conglomerates at Shap Wells (sites 14 and 15 in Figure 2) deposited during the Dinantian sub-System of Early Carboniferous times. It underlies Chadian-age rocks and may date from the Courceyan Stage (Garwood, 1913). Material derived from the Shap Granite is in the form of disaggregated feldspar phenocrysts and could not be reliably sampled for palaeomagnetic study.

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Table 2. Site and group mean palaeomagnetic results from the Shap Granite and country rocks

Site no.	D (*in situ*)	I (*in situ*)	D (tilt-adjusted)	I (tilt-adjusted)	N	R	k	α_{95}
(i) Adamellite intrusion and xenoliths, A components								
1	271·3	5·1	270·1	17·9	11	10·11	11	14·2
2	276·2	12·5	278·2	22·7	7	6·60	15	16·1
3	275·0	−6·8	274·8	3·6	4	3·36	5	47·6
4	275·3	11·5	277·2	21·8	6	5·61	13	19·4
5	277·1	20·4	280·3	30·4	13	12·21	15	11·0
6	268·0	7·1	269·1	18·0	7	6·75	24	12·5
7	278·9	20·5	282·2	30·3	10	9·27	12	14·4
8	286·5	3·2	287·7	12·1	16	15·04	16	9·7
10	278·1	11·1	280·0	21·1	11	10·35	15	12·1
(a) Normal polarity A3 magnetizations								
	92·0	−2·7	92·9	−13·4	38	34·57	11	7·4
(b) Reversed polarity A3 magnetizations								
	279·7	11·2	281·7	21·0	56	51·26	12	5·8
(c) Combined (reversed) mean								
	276·8	7·0	278·2	17·1	94	85·90	12	4·5
(Palaeomagnetic Pole: 263·8°E, 6·8°N, d_p/d_m = 2·3/4·5°)								
(ii) Stockdale Rhyolite in Shap aureole, A3 component								
11	290·3	8·4	292·2	16·7	16	14·13	8	14·1
(iii) Borrowdale Volcanics in Shap aureole, A3 component								
12	285·8	−0·8	286·3	8·3	13	12·23	11	15·5
(iv) Silurian mudstones in Shap aureole								
13	208·2	25·4	203·8	34·8	27	21·55	4	14·3
(v) Lower Carboniferous conglomerate, Shap Wells								
14/15	184·0	42·7	–	–	24	3·85	1	–

D and I are the mean declination and inclination, respectively, derived from component populations recognized in N samples. R is the magnitude of the resultant vector and k is the Fisherian precision parameter (= $(N-1)/(N-R)$). α_{95} is the radius of the cone of 95% confidence about the mean direction; d_p and d_m are the radii of the oval of confidence about the equivalent pole position in the colatitude direction and at right angles to it, respectively. Tilt adjustments are for the tilt of the unconformably overlying Lower Carboniferous sediments (see text); the listed result for the Lower Carboniferous conglomerate is derived from the mean of separate clast directions.

However, 30 clasts of tuffs, lavas and mudstones derived from the adjoining Lower Palaeozoic outcrop were sampled with multiple cores from three clasts.

Remanence components isolated by thermal demagnetization come from 24 clasts, all in volcanic samples derived from the adjoining BVG outcrop. Most magnetizations have two-component structures with good within-clast agreement recognized in the h.b.t. components (samples 5, 13 and 15 in Figure 6); no common overprint is evident. The group mean from the clast lithologies is random at the 95% confidence level (Table 2) and only two clasts (15A, 16) include directions similar to the axis recognized in the underlying pluton; whilst most clasts exhibit dual-component behaviours, in no case is the component structure similar to the remanence in the adamellite or to the comparable BVG lithologies in th aureole. Hence the test implies that remanence isolated from the pluton predates deposition of the conglomerate in Early Carboniferous times.

The overall mean of the A3 components in the pluton (D/I = 277/ + 7°) is related to primary igneous cooling in a reversing field at *c.* 395 Ma and therefore merits correction for the small easterly tilt of the adjoining Lower Carboniferous strata (Figure 2 and Table 2). The tilt-adjusted remanence (D/I = 278/ + 17°) is shallower and of more westerly declination than the Upper Silurian–Lower Devonian

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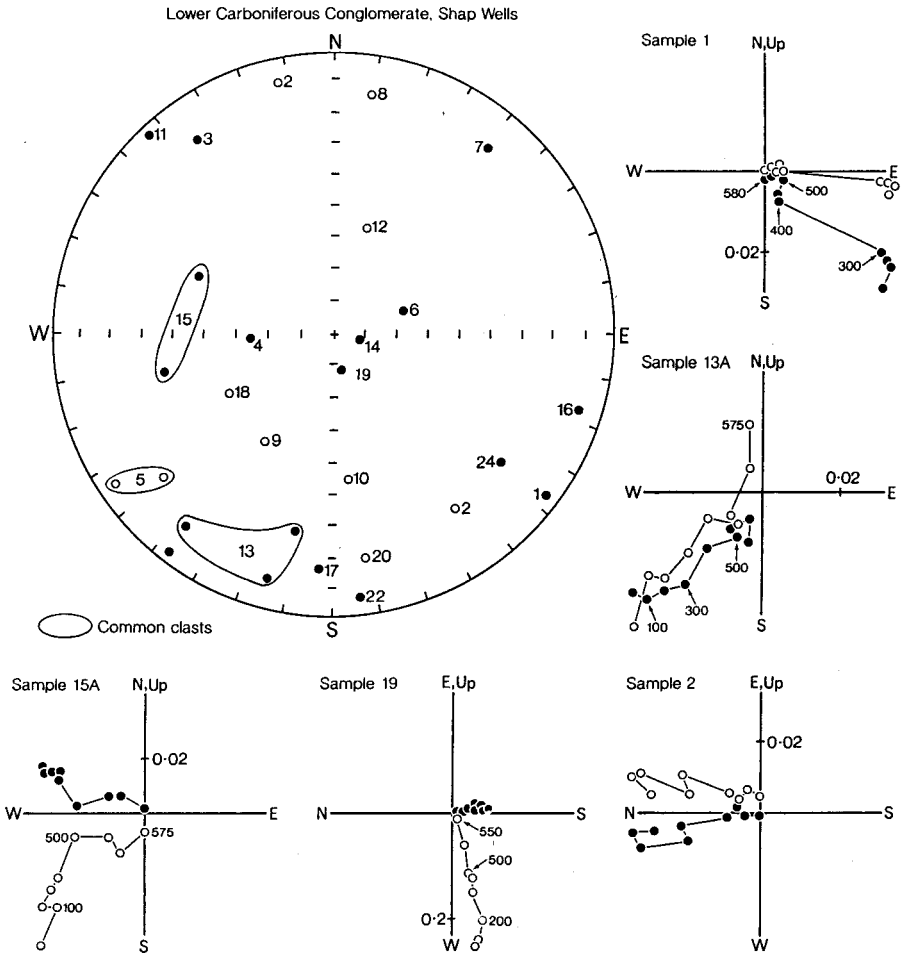


Figure 6. Results of thermal demagnetization of clasts derived from the Borrowdale Volcanic Group in a Lower Carboniferous basal conglomerate at Shap Wells (sites 14 and 15 in Figure 2). The orthogonal plots show projections onto the horizontal plane as closed circles and onto the vertical plane as open circles; axes are scaled in units of intensity $\times 10^{-5}$ A m^2/kg. The equal-area projection shows the distribution of component directions resolved from conglomerate clasts. Directions from the same clast are enclosed; these are averaged for the group calculation in Table 2

directions resolved from regions to the north (Sallomy and Piper, 1973; Torsvik, 1985) and to the south in Britain (Channell *et al.*, 1992b; Setiabudidaya *et al.*, 1994) which identify a typical 'Old Red Sandstone' dual-polarity axis with NE − /SW+ orientation. This axis demonstrably predates Acadian deformation in the Welsh Borderlands (Channell *et al.*, 1992b) whereas the Shap A3 magnetization post-dates all but the latest brittle phases of Acadian deformation in the Lake District. Hence the difference between them probably records APW during Early Devonian times (Section 5).

4b. The Skiddaw Granite aureole

The Skiddaw Granite is another member of the post-orogenic granite suite. It was emplaced into deformed Skiddaw Slates and the Carrock Fell Complex (Figures 1 and 7). Only three small outcrops are exposed in the floor of the Caldew Valley but a wide thermal aureole identifies this as the roof zone of a large subcrop rising from the regional batholith (Eastwood *et al.*, 1968; Figure 1). Illite, developed along regional cleavage in the country rock slates, is dated 395 Ma by K–Ar (Ineson and Mitchell, 1974), comparable to a

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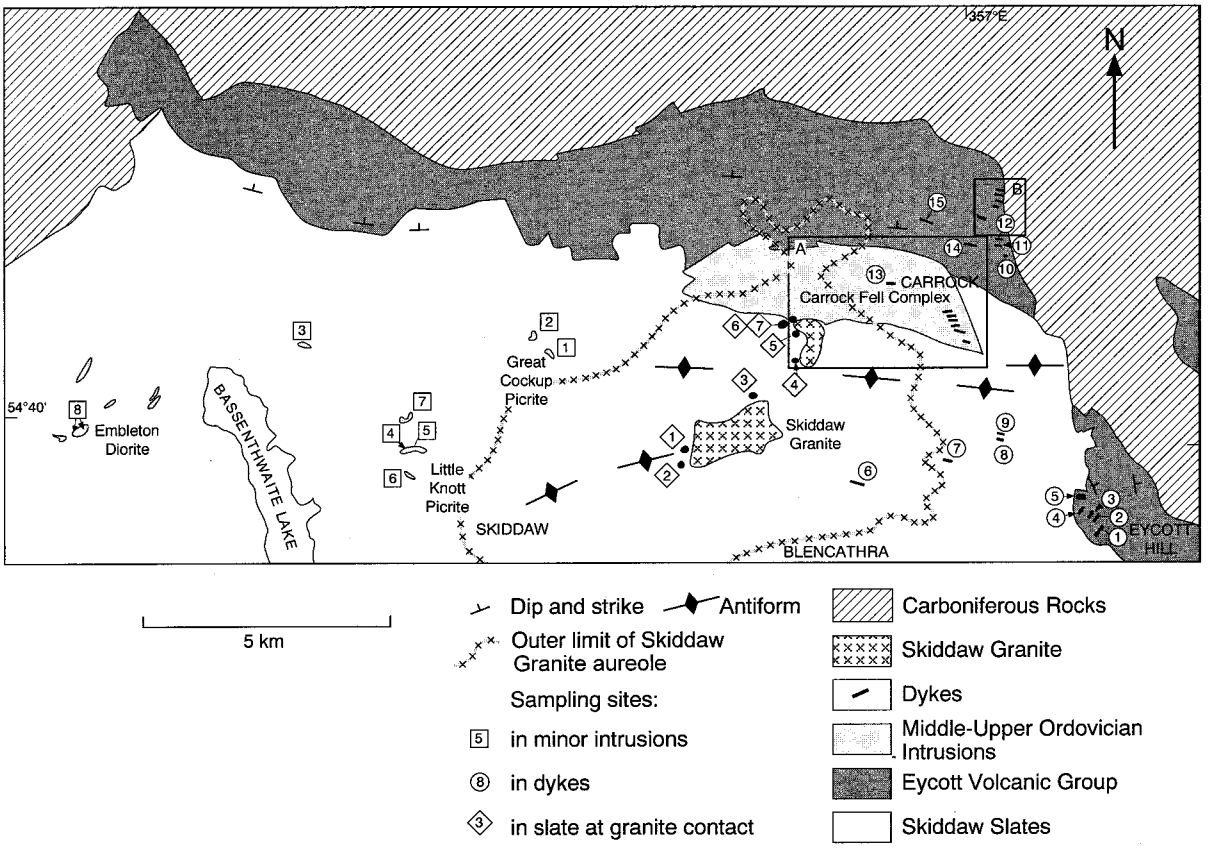


Figure 7. Geological map of the northern part of the Lake District terrane (see Figure 1) simplified after the 1:63 360 Geological Map and showing palaeomagnetic sampling sites of this study. The extent of the thermal aureole of the Skiddaw Granite in the country rocks is indicated together with axes of major F2 folds noted in the text. The boxed areas are (A) the Carrock Fell Complex shown in more detail in Figure 13 and (B) a dolerite dyke swarm studied by earlier investigation (Piper *et al.*, 1978) and referred to in Section 4j

determination on the granite of 397 ± 4 Ma (K–Ar on biotite (Shepherd *et al.*, 1976)). The regional Acadian cleavage affects strata as young as Downtonian (early Gedinnian) in age; granite emplacement overlapped this event to produce a comparable relationship between emplacement and late regional cleavage to that seen in the Shap Granite aureole (Soper and Roberts, 1971). Emplacement ages of the two plutons are thus indistinguishable. At outcrop the Skiddaw pluton is a coarse-grained leucocratic granite which has suffered extensive hydrothermal alteration with partial conversion to a greisen; it is therefore unsuitable for palaeomagnetic study and sampling was concentrated on hornfelsed Skiddaw Slates in the country rock margin at seven sites (diamond symbols 1–7 in Figure 7).

The anisotropy of magnetic susceptibility (AMS) is conveniently expressed in terms of a triaxial ellipsoid with maximum (k_1 or k_{max}), intermediate (k_2 or k_{int}) and minimum (k_3 or k_{min}) axes. The Skiddaw hornfels is derived from a parent Skiddaw Slate but the degree of AMS (k_1/k_3) is mostly less than 1·15 and probably reflects static recrystallization (Figure 8). A broad girdle of k_2 and k_3 axes of oblate fabrics in the contact slates is orientated NNE–SSW with westerly dips (Figure 9); streaking in the k_1 directions records the near E–W strike of the regional F2 fold and exhibits the easterly plunge of this fold structure (Figure 7; Roberts, 1971). Hence recrystallization to hornfels has been dominated by a fabric inherited from the F2 folding with the NNE–SSW girdle resulting from refolding and amplification of this structure during Acadian and Variscan Episodes (Section 4g).

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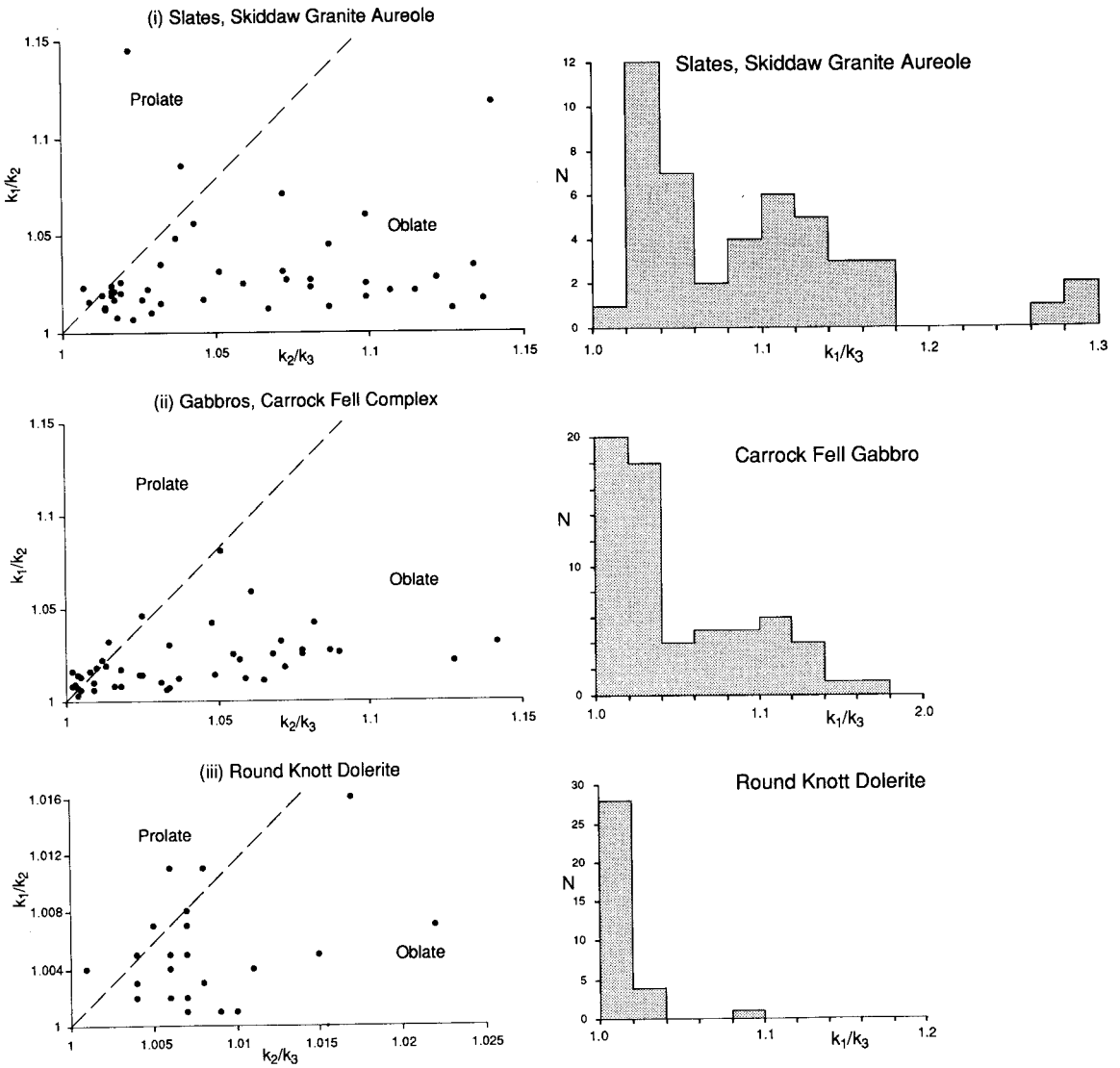


Figure 8. Histograms of anisotropy of susceptibility (AMS) degree (k_1/k_3) in three contrasting rock facies in the Carrock Fell region of the northern Lake District. The plots of k_1/k_3 against k_2/k_3 define the oblate (foliated) or prolate (lineated) shapes of the AMS ellipsoids in each rock facies

Slates in the contact aureole show two types of thermomagnetic spectra (Figure 3). Exponential decrease in J_s is the signature of a dominant paramagnetic component, although some conversion to magnetite takes place on cooling; biotite and chlorite are important constituents of the slates in the granite aureole (Eastwood *et al.*, 1968) and are probably responsible for this signature. In the second curve type an inflection at about 350°C indicates breakdown of a ferromagnetic mineral (probably pyrrhotite) with growth of magnetite identified on the cooling curves.

As a result of this latter effect, most cores at sites 1 to 5 showed little or no systematic component behaviour with progressive thermal demagnetization; l.b.t. components were fully subtracted by 300°C and typical directions close to the present field are attributable to Recent field acquisition. The only evidence for a more ancient component was found in hornfels within a few metres of the granite contact at sites 6 and 7,

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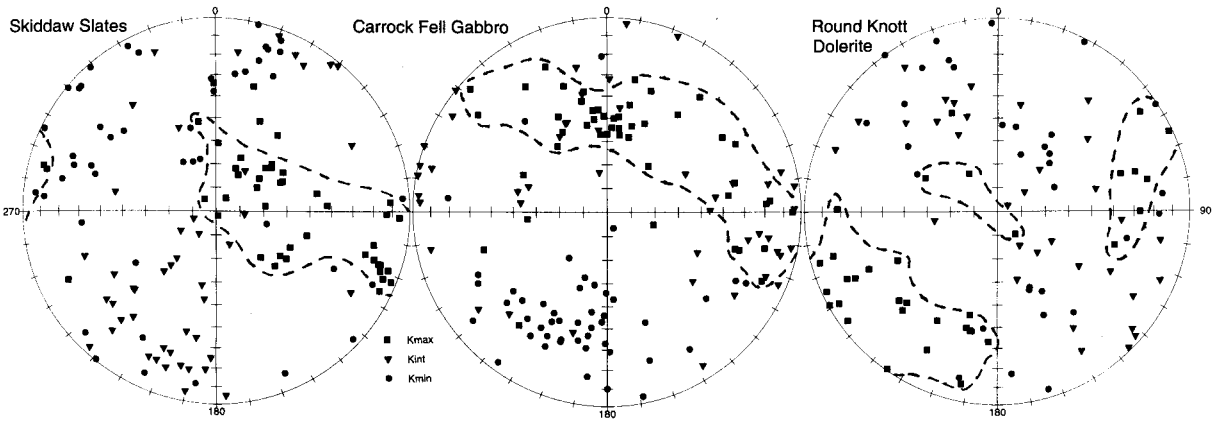


Figure 9. Equal-area lower-hemisphere projections showing *in-situ* directions of the maximum (k_{max}, squares), intermediate (k_{int}, inverted triangles) and minimum (k_{min}, circles) axes of AMS ellipsoids in the Skiddaw Slates of the Skiddaw Granite aureole, the Layered Gabbro Series of the Carrock Fell Complex and the Round Knott Dolerite of the Carrock Fell Complex. The dashed lines embrace the main distributions of k_{max} axes

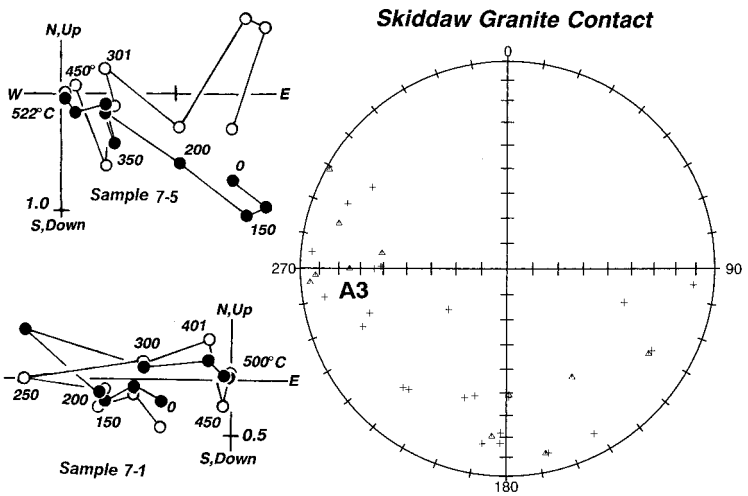


Figure 10. Typical progressive thermal demagnetization results (*in situ*) on samples of hornfels from the aureole at the Skiddaw Granite at sites 6 and 7 (other sites shown in Figure 7 yielded only present field components). Symbols used in the orthogonal projections are as for Figure 4. The equal-area projection shows directions (*in situ*) of non-viscous components of magnetization resolved from hornfels in the Skiddaw aureole; the majority of these components come from sites 6 and 7 within a few metres of the granite contact

where poorly stable components with intermediate unblocking temperatures and shallow inclination are subtracted by 450–500°C, appreciably below the Curie point of magnetite. Since both antiparallel easterly and westerly directions are present (samples 7-1 and 7-5 in Figure 10) and this axis is close to the A3 magnetization in the contemporaneous Shap adamellite, it probably records a thermoremanent magnetization acquired during emplacement of the adjoining pluton.

The distribution of non-viscous component directions in Figure 10 is a weak record of the A3 dual polarity recognized in the Shap Granite and aureole; some scattered southerly directions are probably later Palaeozoic acquisitions. The mean direction derived from the 15 slate samples of the former population (four reversed with respect to the remainder) is $D/I = 282/+6°$ following correction for tilt of the Carboniferous outcrop to the east (Table 3). Recognition of the A3 axis within, or adjoining, the Shap and Skiddaw bodies constrains the palaeofield direction during post-emplacement cooling of the regional batholith.

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Table 3. Palaeomagnetic results from hornfels derived from Skiddaw Slate in the aureole of the Skiddaw Granite

	D (*in-situ*)	I	D (tilt-adjusted)	I	R	k	α_{95}
(i) Mean, A3 components							
15 samples	281·9	−1·1	282·1	6·1	13·25	8	14·4
(Palaeomagnetic Pole: 258·8°E, 9·5°N, d_p/d_m = 7·2/14·4°)							
(ii) Mean SW+ components							
8 samples	205·9	35·9	201·1	46·6	7·19	9	20·0
(Palaeomagnetic Pole: 338·5°E, 46·6°N, d_p/d_m = 13·2/23·2°)							

Symbols are as for Table 2

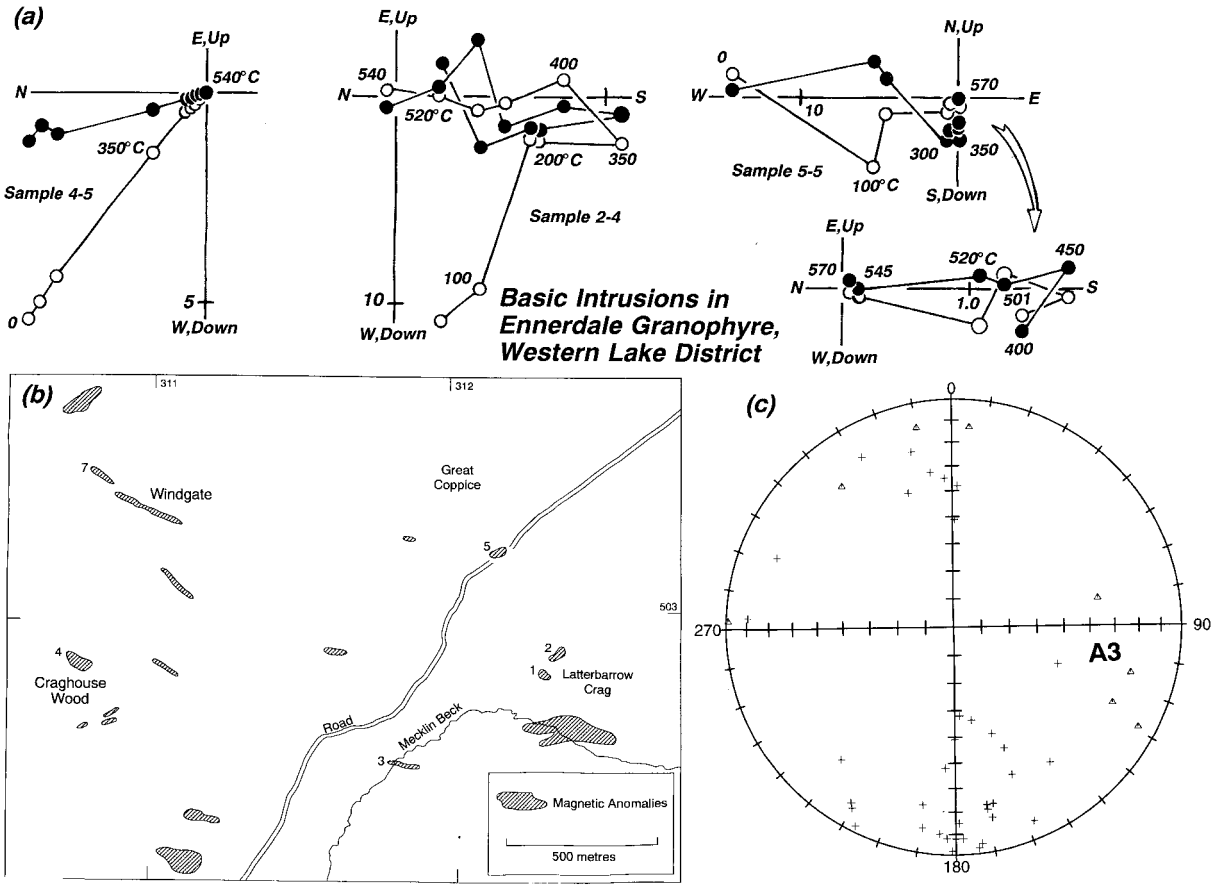


Figure 11. (a) Examples of progressive thermal demagnetization of samples from basic bodies in the Ennerdale Granophyre; symbols are as for Figure 4. (b) Outline map of the southern part of the Ennerdale Granophyre (see Figure 1 for location) showing the distribution of magnetic anomalies within the leucocratic granophyre and palaeomagnetic sampling sites. (c) Equal-area projection showing directions (*in situ*) of non-viscous components of magnetization resolved from these bodies

4c. The Ennerdale Granophyre

The Ennerdale Granophyre is emplaced in the western Lake District as a sub-volcanic pluton near the base of the BVG (Figure 1). It is intruded by the Eskdale Granite which comprises a large outcrop immediately to the south of the study area (Figure 11) where it is also the major surface expression of the regional batholith

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in the western Lake District (Figure 1). A Rb–Sr determination of 420 ± 4 Ma (Rundle, 1979) on the granophyre is younger than a determination on the Eskdale Granite (429 ± 4 Ma); it probably reflects problems of dating fine-grained acid igneous rocks by Rb–Sr and is likely to be a minimum estimate (O'Brien *et al.*, 1985).

The granophyre is primarily an intergrowth of perthite and quartz with only minor amounts of mafic accessories altered to chlorite and epidote. However, small dolerite intrusions are associated with basified and hybrid granophyres (Firman, 1978). Although poorly exposed, they are more strongly magnetic than the host granophyre and can be mapped from ground magnetic surveys. These identify impersistent WNW–ESE trending anomalies; outcrop was detected and sampled at seven of these anomalies (sites 1-5 and 7 in Figure 11b plus an additional site (6) near Wastwater Lake to the east). Whilst intrusion ages are uncertain, an origin related to emplacement with, or soon after, the granophyre is probable because they are not observed in the younger Eskdale Granite. Site 7 is in basic facies of the granophyre pluton. Site 6 near Wastwater Lake may have a different origin because a dyke-like form is suggested by a long linear outcrop with discrete contacts.

ChRMs are dominated by components with southerly directions and shallow to intermediate positive inclination recovered following removal of present field components (sample 2-4 in Figure 11a) or a shallow A3 component (sample 5-5) recording overprinting by the regional batholith (Sections 4a and 4b). The exception is site 4 (e.g. sample 4-5 in Figure 11a) where single and very stable NNW positive components are present. The population of southerly directed magnetizations with three antiparallel components (Figure 11c) has a mean of $D/I = 175/+26^\circ$ (Table 4). Aberrant site 4 has a magnetization comparable to the Carrock dyke swarm in the eastern Lake District (Section 4j).

Table 4. Site and group mean palaeomagnetic results from basic facies and dolerites in the Ennerdale Granophyre

Site no.	Grid ref.	D	I	N/n	R	k	α_{95}
1	3122 5028	–	–	7/0	–	–	–
2	3122 5029	170·5	23·0	10/8	7·60	18	13·5
3	3119 5025	182·2	12·8	7/6	5·87	35	11·5
4	3108 5029	349·6	34·9	7/7	6·82	33	10·6
5	3121 5032	191·4	7·6	6/4	3·85	20	21·2
6	3139 5055	151·9	51·6	8/7	6·54	13	17·3
7	3108 5036	172·2	26·8	10/6	5·56	11	20·8
(i) Group mean, S + /N− components							
32 samples		174·9	25·8		28·69	9·4	8·8
(Palaeomagnetic Pole: 2·0°E, 21·9°S, $d_p/d_m = 5{\cdot}1/9{\cdot}5^\circ$)							
(ii) Devonian (A3) overprints							
8 samples		285·0	18·6		7·69	22·4	12·0
(Palaeomagnetic Pole: 260·1°E, 16·5°N, $d_p/d_m = 6{\cdot}5/12{\cdot}4^\circ$)							

The National Grid Square is SD. Other symbols as for Table 2.

Since the granophyre is intruded by the large Eskdale Granite pluton which attains the surface near the southern part of the study area (Figure 1), pre-Eskdale bodies in the granophyre are likely to record a cooling remanence related to granite emplacement. The Eskdale Granite is dated by Rb–Sr and K–Ar (420 ± 4 and 427 ± 8 Ma (Rundle, 1979)) and is considered to be syntectonic. Tilt adjustments are therefore uncertain but are unlikely to be appreciable both because the body is part of a large and massive outcrop more than 15 km N–S in width, and because local tilts of the BVG are small. The Palaeolatitude derived from the *in-situ* result (13°S) is close to data from late Llandovery sediments in the Howgill Fells immediately to the east of the Lake District terrane (12°S (Channell *et al.*, 1993)) and from mid-Wenlock rocks of Britain close to the Variscan Front (12°S (Torsvik *et al.*, 1993)); this is compatible with the radiometric age of 429 Ma.

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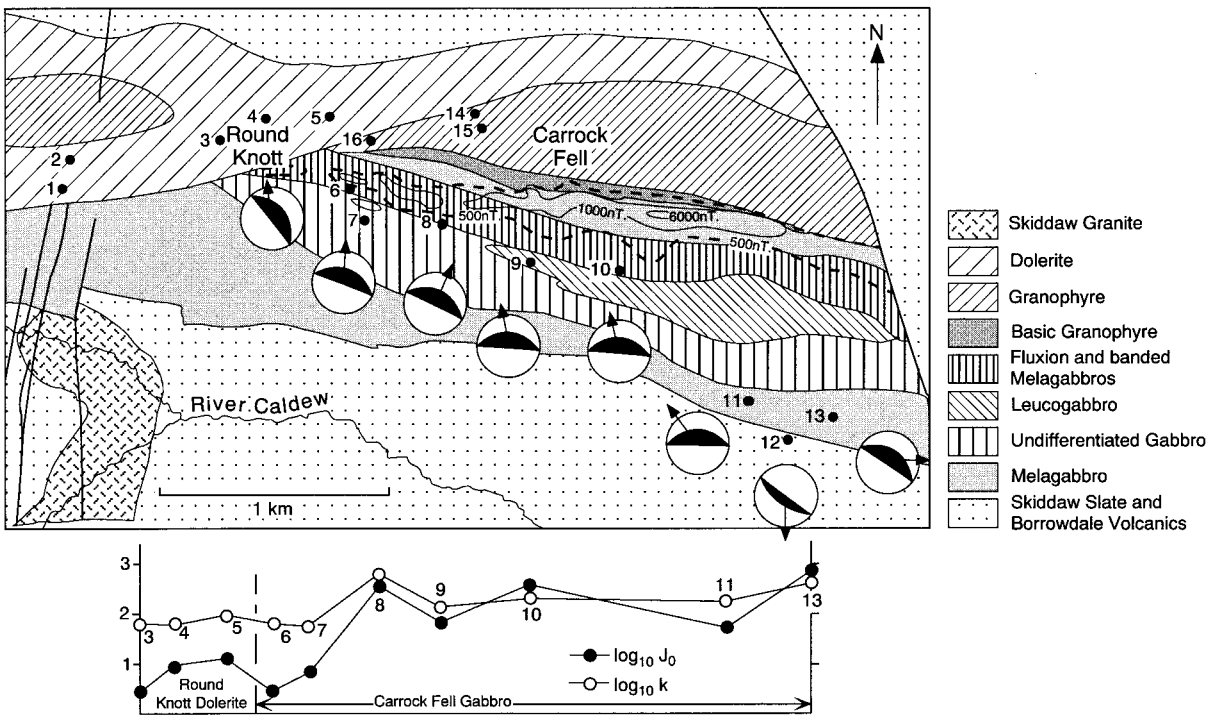


Figure 12. Geological map of the Carrock Fell Complex simplified after Eastwood *et al.* (1968), showing the distribution of palaeomagnetic sampling sites of this study. Also shown are the orientations of AMS ellipsoids in the Layered Gabbro Series and gabbro hornfels; the arrows point in the k_1 direction and the sectors show the orientations of the foliation plane incorporating k_1 and k_2

4d. The Carrock Fell Complex: the Layered Gabbro Series

The Carrock Fell Complex comprises a Gabbroic Series emplaced along the contact between the Skiddaw Slates and Eycott Volcanic Group (Figures 1 and 7). It possesses a broad symmetry with iron-ore and pyroxene/hornblende-rich melagabbros at the margins of the body giving way inwards to banded gabbros displaying alternate mafic and felsic layers, and then to feldspathic leucogabbros which sandwich a central mesocratic gabbro (Eastwood *et al.*, 1968; Firman, 1978; Bowden and Hunter, 1990; Figure 12). A high-amplitude magnetic anomaly reflects the general strike of the layering in the unexposed vicinity of the contact with the Carrock Fell Granophyre. None of the exposed gabbro facies are strongly magnetized in a way which can explain this anomaly (Figure 12) and it is probably caused by buried segregations of pyrrhotite within the gabbro (also found located near the base of the complex in its westerly extension (see Patrick, 1980)) or by strongly magnetized ferrogabbros.

The igneous banding has steep northerly dips and all workers, until recently, have considered that emplacement occurred as a sub-vertical body in the present orientation (Harker, 1894; Hollingworth, 1938; Eastwood *et al.*, 1968; Briden and Morris, 1973). Harris and Dagger (1987), however, evaluated field evidence to show that the layered gabbros were originally emplaced as part of a sub-horizontal tabular body and folded by the regional F2 deformation. Specifically, these authors show that the distribution and orientation of country rock xenoliths is compatible with a former floor sited at the present southern contact (Figure 12); rhythmic layering and modal distributions in the banded gabbros support the same conclusion.

K–Ar biotite dates from the Layered Gabbro Series yield a range of ages between 439 ± 10 and 468 ± 10 Ma (Rundle, 1979, 1981), indicating a mid-to Late Ordovician age. Geochemical evidence for a common tholeiitic fractionation trend (Hunter, 1980; O'Brien *et al.*, 1985; Bowden and Hunter, 1990)

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supports a correlation with the adjoining EVG and supports the view that the Gabbro Series is part of a subvolcanic complex which fed the lavas above (Eastwood *et al.*, 1968).

The Layered Gabbros have weak, typically oblate, AMS fabrics with anisotropy degrees (k_1/k_3) mostly less than 1·04 but sometimes higher than 1·10 (Figure 8). The k_1 and k_2 axes occupy a girdle dipping northwards at intermediate angles (Figure 9) similar to, but typically slightly less than, the dips of the mineral layering observed at the eastern margin of the outcrop (50–70°N (Eastwood *et al.*, 1968)); k_3 axes dip at intermediate angles to the south, perpendicular to this foliation. Fabric in the magnetic susceptibility therefore approximately records the primary igneous layering in the body and contrasts with fabric in the adjoining country rock hornfels (site 12, Figure 12). However, the k_1 axes also dip mostly to the north at intermediate angles (Figure 9) and close to the dip of the F2 fold limb; they therefore identify a relationship with the regional tectonic history and suggest that lineation has been imparted to the igneous foliation by the F2 deformation (Section 4g). Thermomagnetic spectra in the gabbros are dominated by strong saturation remanences with Curie points close to that of pure magnetite and little alteration on heating (Figure 3).

The Layered Gabbros of both melanocratic and leucocratic facies have strong and very stable magnetizations; single components are isolated above 200°C and dominated by narrow and discrete ChRMs typically subtracted over 20°C or less between 540 and 560°C (e.g. samples 6-1, 8-5 and 10-1 in Figure 13). The majority of directions have northerly declinations with shallow negative inclinations (mean $D/I = 357/-16°$, Figure 14). The exception is a single site (9) where a majority of cores (e.g. sample 9-4 in Figure 13) are overprinted by a field direction recorded in late dolerite dykes cutting the complex (Section 4j).

The dominance of one magnetite-held component in the remanence structure of the Layered Gabbros accounts for good agreement between this study and the earlier result derived from a.f. demagnetization and endpoint analysis ($D/I = 353/-18°$ (Briden and Morris, 1973; Morris, 1973)). Skiddaw Slate at the lower margin of the complex has been baked to a white hornfels; five of eight samples from a site (12) within 4 m of this contact yield discrete magnetite-held components (sample 12-1 in Figure 13) close to the gabbro remanence. Although blocking-temperature spectra are somewhat lower and more distributed than in the gabbro, directions are indistinguishable; behaviours contrast with slate in the granite aureole (Section 4b) and record thermal overprinting by the gabbro complex.

Three episodes of folding (F1–F3) have been recognized in the Skiddaw Slate Group immediately to the south of the Carrock Fell Complex by Roberts (1971). The Layered Gabbros are in the northern limb of a large regional-scale F2 fold with ENE–WSW trending axis (Figure 7). Tight upright folds of the F1 generation are only observed in the Skiddaw Slates (Helm and Roberts, 1971; Wadge, 1971) and are now regarded as soft-sediment deformation. The outcrop of the EVG in Figures 1 and 7 highlights the form of the F2 structure evident from the 1 : 63 360 geological map of the region (Sheet 23, Cockermouth; Eastwood *et al.*, 1968) and identifies a plunge to the east (*c.* 38°/86°E). It has a composite origin: the late Caledonian Acadian F3 folding produced E–W-trending open folds (Roberts, 1971) and was probably responsible for amplifying the F2 structure. In addition, the unconformably overlying Lower Carboniferous rocks dip at gentle angles (typically 10–15°) northwards and eastwards around this margin of the inlier, showing that Variscan deformation has further enhanced the observed amplitude and plunge. Since successive compressive events were directed approximately N–S, the structure can be treated as a single entity for purposes of tilt adjustment (Harris and Dagger, 1987).

These latter authors note that adjustment for plunge and residual tilt of the a.f. demagnetized result from the Carrock Complex (Briden and Morris, 1973) moves the *in-situ* result ($D/I = 353/-18°$) to a steeper negative inclination ($D/I = 315/-58°$), more compatible with results from the EVG ($D/I = 5/-43°$) and the BVG ($D/I = 341/-47°$ and $D/I = 347/-62°$; see discussion in Section 5 and in Briden and Morris (1973), Faller *et al.* (1977), Channel and McCabe (1992) and Piper *et al.*, (1997). This correction also bring poles to the layering in the Gabbro Series near to the vertical (Harris and Dagger, 1987), supporting emplacement (and magnetization) of the Layered Gabbro Series as a quasi-horizontal sheet.

The AMS data and new palaeomagnetic results permit refinement to this result. The magnetic fabric defines a foliation plane dipping to the north with a mean orientation direction of $D/I = 14/46°$.

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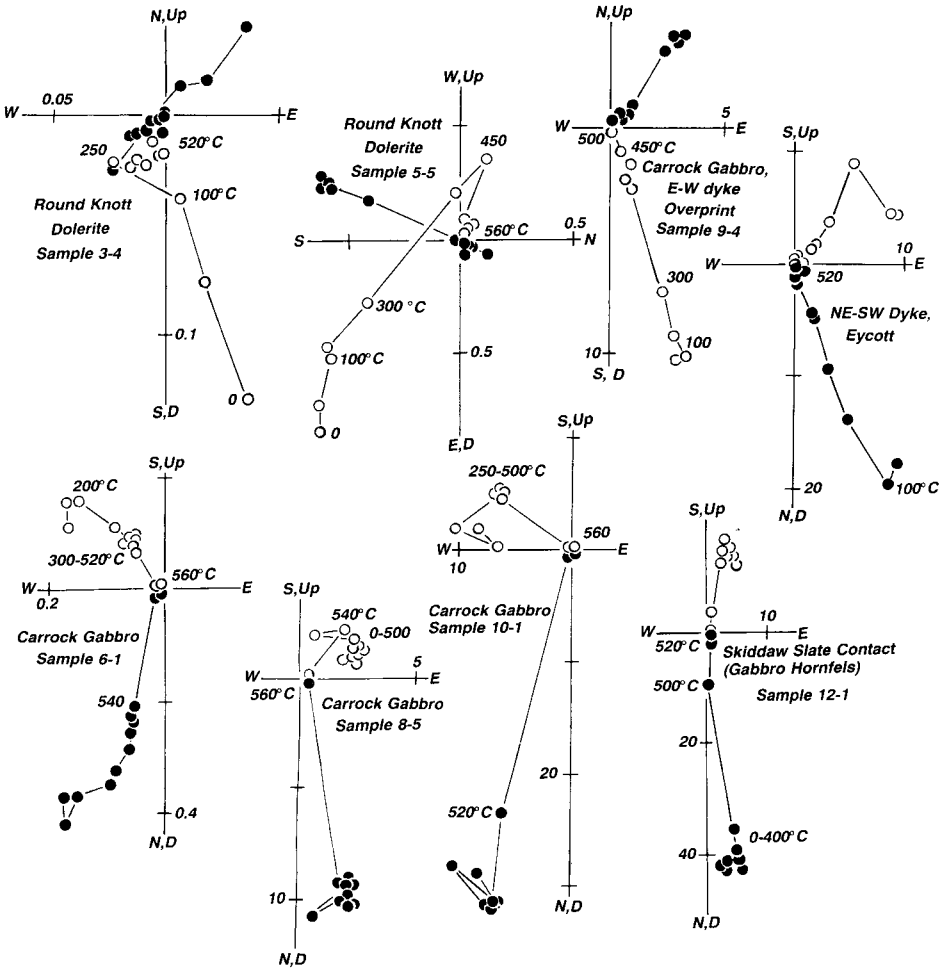


Figure 13. Results of progressive thermal demagnetization (*in situ*) of samples from the Carrock Fell Layered Gabbro Series, contact of the gabbros with the Skiddaw Slate (site 12), and Round Knott Dolerite intrusion. The results of progressive treatment are shown as orthogonal projections onto the horizontal plane (solid circles) and vertical plane (open circles); the figures are magnetizations in $\times 10^{-5}$ A m^2/kg

(Significant deviation occurs at site 13 near the eastern margin of the complex (Table 5); since this is also sited close to a possible boundary fault (Figure 12) it is excluded from the fitted planes.) Two-stage tilt adjustment then moves the *in-situ* direction to $D/I = 17/-58^\circ$ (Table 6). This is close to the plunge and tilt-adjusted mean from the EVG ($D/I = 5{\cdot}0/-43{\cdot}3^\circ$ (Piper *et al.*, 1997)) and reduces the residual declination anomaly in the earlier result (Harris and Dagger, 1987; Figure 15). The inclination lies between the EVG/BVG primary magnetizations and the regional Late Ordovician overprint ($D/I = 347/-62^\circ$) on the BVG and is therefore compatible with magnetization late in the EVG/BVG igneous episode.

4e. The Carrock Fell Complex: the Carrock Granophyre

The Layered Gabbros of the Carrock Fell Complex are intruded by two igneous bodies, the Carrock Fell Granophyre and the Round Knott Dolerite (Eastwood *et al.*, 1968). The granophyre is a typical sub-volcanic intrusion emplaced as a steep-sided body cutting the inclined layering of the gabbros. Geochemical data (Hunter, 1980; O'Brien *et al.*, 1985) relate it to a tholeiitic trend in common with the gabbros and EVG.

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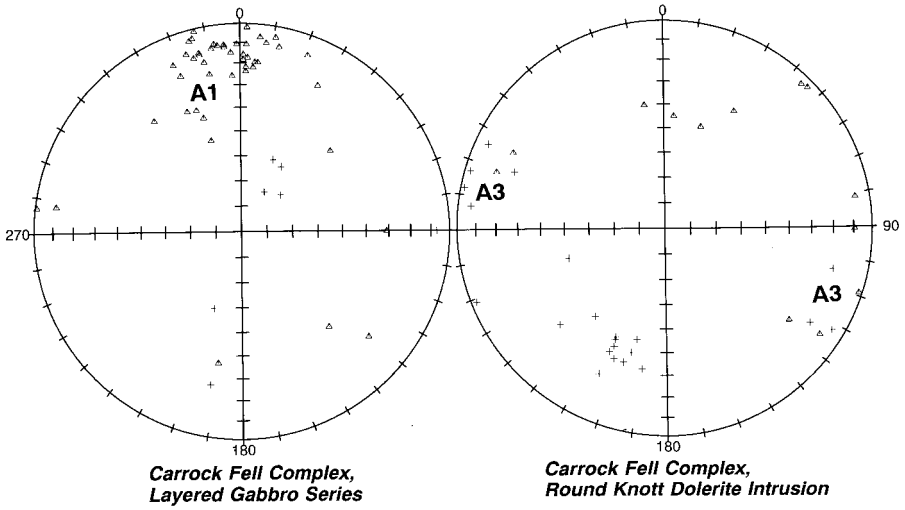


Figure 14. Directions of (non-viscous) components of magnetization (*in situ*) in the Carrock Fell Layered Gabbro Series and the Round Knott Dolerite intrusion; equal-area projection

However, the granophyre has yielded a younger Siluro-Devonian radiometric age: Rundle (1979) quotes a 416 ± 20 Ma Rb–Sr determination but notes the high error and possible influence by the Skiddaw Granite. As with the Ennerdale Granophyre, this result probably reflects the difficulties of dating fine-grained acid rocks by Rb–Sr; O'Brien *et al*. (1985) advocate a close geochemical link between the granophyre and the ferrogabbros of the Carrock Fell Complex and hence a Middle to Late Ordovician age.

The granophyre forms the summit of Carrock Fell and initial indications of strong lightning-induced isothermal remanent magnetizations (IRMs) necessitated a sampling strategy of distributed coring across exposure in the vicinity of sites 14–16 in Figure 12. In a collection of 51 cores, 11 yielded strong

Table 5. *In-situ* orientations of mean AMS ellipsoids in the Carrock Fell Complex and Skiddaw Hornfels

Site	N	k_1	k_2	k_3
Carrock Fell Gabbro				
6	5	353/20	98/33	191/27
7	7	359/53	94/3	185/37
8	7	21/39	289/2	196/51
9	5	346/42	84/6	178/38
10	7	350/37	85/23	206/35
11	4	305/49	91/29	195/24
13	8	93/28	341/37	210/42
Skiddaw Slates				
1	3	177/76	19/10	289/3
2	11	28/55	182/32	280/12
3	2	98/12	192/21	339/65
4	5	88/5	180/29	352/60
5	10	108/5	218/77	18/12
6	11	55/61	218/18	311/8
7	7	130/67	23/7	290/21

N is the number of measurements used to derive the best fitting ellipsoid and the directions of the maximum (k_1), intermediate (k_2) and minimum (k_3) axes are given in terms of declination (°E)/inclination plotting on the lower hemisphere. Site numbers refer to Figure 12 (Carrock Fell Gabbro) and Figure 7 (Skiddaw Slates)

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Table 6. Site and group mean palaeomagnetic results, Carrock Fell Complex

Site no.	N/n	D (*in-situ*)	I (*in-situ*)	D (tilt-adjusted)	I (tilt-adjusted)	R	k	α_{95}
(i) Carrock Fell Gabbros								
6	9/8	353·6	−14·0	6·9	−55·7	7·83	40·1	8·9
7	8/4*	356·6	−30·8	14·7	−72·4	3·70	10·1	30·4
8	8/6	1·2	−16·3	20·7	−57·6	5·89	45·5	10·0
9	7/2	334·4	−20·4	331·3	−57·9	1·91	10·8	–
	7/5†	38·4	65·6	41·3	53·9	4·96	97·6	7·8
10	8/6	354·7	−13·6	8·9	−55·4	5·94	89·3	7·1
11	7/6	353·7	−11·2	7·3	−52·9	5·93	72·5	7·9
12	8/5	7·1	−17·4	31·3	−57·4	4·95	81·5	8·5
14	8/4*	‡273·5	3·1	—	—	3·84	18·8	21·8
15	9/4	339·9	−43·0	300·5	−78·7	3·98	125·3	8·2
Group mean								
7 sites		354·5	−17·9	13·3	−59·8	6·88	49·3	8·7
(Palaeomagnetic Pole: 346·9°E, 5·9°N, d_p/d_m = 14·7/20·9°)								
36 samples		356·9	−16·4	17·4	−58·1	34·62	25·4	4·8
(Palaeomagnetic Pole: 343·5°E, 4·6°N, d_p/d_m = 2·6/6·0°)								
(ii) Carrock Fell Granophyre								
29 samples		354·3	−64·4	319·0	−67·1	26·6	11·7	8·2
(Palaeomagnetic Pole: 23·8°E, 20·0°N, d_p/d_m = 10·5/13·1°)								
(iii) Round Knott Dolerite								
1	17/3†	290·2	0·3	290·7	6·1	2·88	16·6	31·3
2	8/4†	103·2	−0·9	103·8	−7·9	3·65	8·6	33·2
3	10/8	201·2	37·9	195·4	48·1	7·93	100·9	5·5
4	9/5‡	105·4	−8·3	107·3	−14·8	4·90	39·1	12·4
	9/3	226·6	30·0	226·0	41·8	2·75	7·9	47·1
5	9/6	195·4	41·9	187·5	51·3	5·84	30·3	12·4
	9/2‡	299·9	−6·1	298·9	−2·0	1·95	20·3	–
(a) NNE − /SSW+ components								
21 samples		205·9	37·5	200·5	48·2	20·00	20·1	7·3
(Palaeomagnetic Pole: 338·9°E, 4·3°S, d_p/d_m = 5·0/8·6°)								
(b) A3 (Devonian overprints)								
13 samples		288·7	2·7	289·5	8·7	12·20	15·0	11·1
(Palaeomagnetic Pole: 253·4°E, 14·8°S, d_p/d_m = 5·6/11·1°)								

*Two antiparallel components included in sit mean calculation.
†Overprinted by regional dyke swarm.
‡Interpreted as A3 (Devonian) overprints.
Symbols as for Table 2.

magnetizations which, although showing convergent components, had random directions; these are interpreted as secondary IRMs and are excluded from further consideration. Remaining samples are between one and two orders of magnitude more weakly magnetized and possess magnetizations mostly with two- and three-component structures (Figure 16). A small number of easterly and westerly directed magnetizations present as either single (sample 40 in Figure 16) or l.b.t. (sample 26) components are readily identified as overprinting effects of the Skiddaw Granite emplaced into the base of the complex.

The most prominent features of the magnetization are components with N–NNW negative or S–SSE positive directions; these appear to record a dual-polarity axis and a reversing field is further suggested by demagnetization behaviour in some individual samples (sample 17 in Figure 16). The northerly magnetizations sometimes include an l.b.t. steeper negative component and an h.b.t. shallower component with

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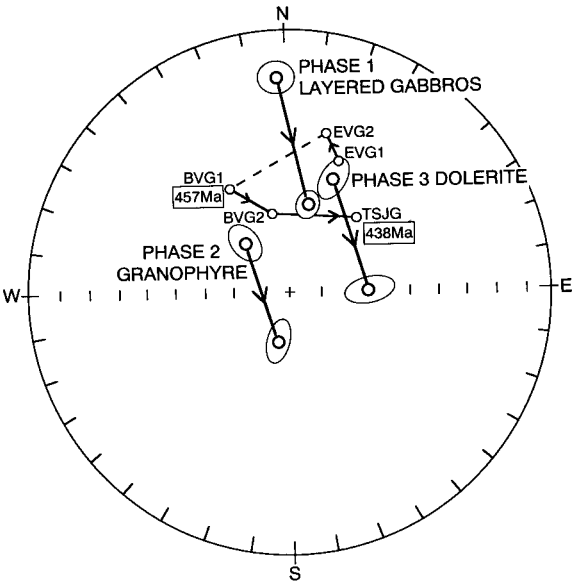


Figure 15. Group-mean directions of magnetizations and α_{95} confidence circles derived from the three successive phases of the Carrock Fell Igneous Complex before and after adjustment for regional tilt. Comparative Upper Ordovician magnetization directions from this terrane also plotted in sequence of age are labelled: EVG1, Eycott Volcanics, Binsey Group; EVG2, Ecyott Volcanics, High Ireby Group; BVG1, Borrowdale Volcanic Group, mean of lava magnetizations; BVG2, Borrowdale Volcanic Group, overprint; TSJG, Threlkeld–St John's Microgranite. Magnetizations of normal polarity are shown in each case. Note that the Layered Gabbro (the first phase of the Carrock Complex) requires correction for the effects of F2 deformation to conform most closely to the comparative results whereas the subsequent phases conform with these results most closely before adjustment

more northerly direction (sample 3 in Figure 16). It is not possible to separate these components in the overall collection and all 29 components belonging to the NNW − /SSE+ population (Figure 16) are combined in Table 6 to yield a mean $D/I = 345{\cdot}3/-64{\cdot}4^\circ$ ($\alpha_{95} = 8{\cdot}2^\circ$). The angle between the normal and reversed axes is $8{\cdot}9^\circ$ which is less than the critical angle ($15{\cdot}2^\circ$) yielding a positive reversal test (McFadden and McElhinny, 1990). Although this is a positive class C test, the component structures and dispersion indicate a complex history of magnetization in Late Ordovician times, although this axis does not correspond to later palaeofield directions from Britain. Adjustment for tilt of the Carboniferous strata nearby moves the mean direction to $D/I = 319/-67^\circ$.

4f. The Carrock Fell Complex: the Round Knott Dolerite

This dyke-like body cuts the Layered Gabbros (Figure 12) and attains a maximum thickness of 0·5 km. Although it appears to divide a once-continuous granophyre body (Figure 12), age relationships between the dolerite and the Carrock Fell Granophyre are unclear (Eastwood *et al*., 1968); the dolerite is probably mostly younger, although any age differences are likely to be slight because the dolerite is locally veined by granophyre. It is, however, much more altered than both the Layered Gabbros and Granophyre, and the younger dolerite dykes, and is referred to as 'diabase' in the early literature (Harker, 1894). Feldspar is profoundly sericitized, whilst widespread development of chlorite and secondary amphibole, as well as peripheral alteration of opaques, are evident in thin section; the dominance of secondary minerals is reflected in a high content of combined water in the chemical analyses (Eastwood *et al*., 1968).

The dolerite possesses only slight magnetic fabric, with anisotropy degrees mostly *c*. 1.01–1.02 (Figure 8). The k_1 axes exhibit a weak tendency to lie in the plane of this steep dyke-like intrusion (Figure 9) cutting the gabbro fabric obliquely and correlating with the ENE–WSW outcrop (Figure 12). Although apparently related to primary emplacement, this fabric probably records alignment of pseudomorphing secondary

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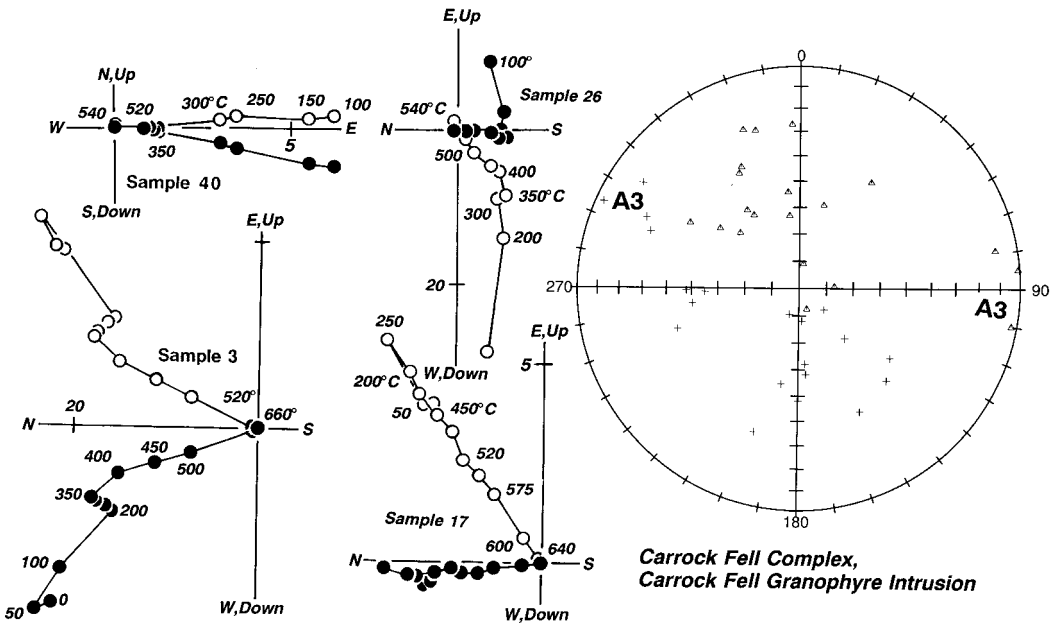


Figure 16. Results of progressive thermal demagnetization of samples of the Carrock Fell Granophyre illustrated *in situ* as orthogonal projections and using the same symbols as in Figure 4. The equal-area projection shows the distribution of component directions (*in situ*) in the granophyre excluding probable IRM components

minerals such as chlorite because thermomagnetic spectra have a dominant paramagnetic expression (Figure 3); weak magnetite Curie points are sometimes evident, with conversion to magnetite apparent in the cooling curves.

In common with earlier observations by Morris (1973) and Briden and Morris (1973), no stable demagnetization behaviour was recognized close to the intrusive contact with the Skiddaw Granite (sites 1 and 2 in Figure 12). Where strong magnetizations are present they are largely unblocked in the first stages of treatment, mostly below 200°C, and are likely to reside in secondary multidomain carriers grown during thermal metamorphism. At greater distances from the granite contact components defined by thermal demagnetization reflect the overprinting effects of the granite and are more prominent in this body than in the Layered Gabbros or Granophyre (Figure 14). Other samples have a SSW+ ChRM following removal of a present field component (sample 3-4 in Figure 13). This is magnetite-held like the granite overprints and, since it is also close to the Early Carboniferous palaeofield direction, cannot positively be identified as older than the granite. However, overprinting of this age is not recognized elsewhere in the region; furthermore four samples illustrate a reversal in this component (sample 5-5 in Figure 13). The angle between axes defined by the normal and reversed components is 11·0°, which is less than the critical angle of 15·2° and yields a positive class C reversal test (McFadden and McElhinny, 1990). Thus in addition to partial overprinting by the Skiddaw Granite, a NNE−/SSW+ axis of magnetization is recognized (Figure 14) with a mean direction $D/I = 206/38°$ (Table 6).

This palaeofield direction is divergent from magnetizations identified in the Layered Gabbros and Granophyre. In view of the profound hydrothermal alteration of the dolerite compared with the other facies of the igneous complex, it is likely to be younger in age and is probably related to pneumatolitic alteration during emplacement of the dolerite. Adjustment for tilt of the Layered Gabbros moves the mean to $D/I = 267/62°$; this would imply that the body had a sheet-like form inclined to the north at the time of magnetization and is clearly inappropriate. Only adjustment for tilt of Carboniferous strata to the east is therefore applied in Table 6.

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4g. The palaeomagnetic record in the Carrock Fell Complex

Application of tilt adjustment implied by the fabric in the gabbro to the Carrock Fell Granophyre moves the remanence to $D/I = 193/-73°$. The *in-situ* remanence in the granophyre is therefore significantly closer to that in the dipping Layered Gabbro series *after* tilt adjustment of the latter (Figure 15); it also then accords most closely with the palaeofield directions in the stratified rocks (EVG and BVG) which are directly referable to a palaeohorizontal. The palaeomagnetic data therefore imply that the granophyre was emplaced after formation of the major fold structure; this is also implicit in its near-vertical orientation. A few steeper NNW- partial l.b.t. overprints in gabbro samples (Figure 14) might record partial overprinting by the granophyre body, whereas no northerly shallow negative *in-situ* remanences like those characterizing the gabbro are observed in the granophyre.

Hence palaeomagnetic evidence identifies the F2 deformation as an event occurring during emplacement of the Carrock Fell Complex and prior to mid-Caradocian times. Although an end-Silurian (Acadian) age for this folding has been proposed on the basis of fold trends (Roberts, 1971) and generally accepted subsequently (Soper and Moseley, 1978), an Upper Ordovician age is supported by several observations: (i) the near-vertical orientations of the Carrock Fell Granophyre and Round Knott Dolerite bodies; (ii) the *in-situ* remanence in the Granophyre is closest to that of the Layered Gabbros after the latter have been adjusted for tilt; (iii) the AMS fabric in the Layered Gabbros, which is primarily related to igneous layering, also records the imprint of F2 deformation whilst no vestige of this episode is evident in the Round Knott Dolerite emplaced late in the igneous cycle (Figure 9); (iv) a pre-Silurian dyke swarm in this region cuts the Carrock Fell Complex and is probably a late expression of the Ordovician magmatism (Eastwood *et al.*, 1968); the dykes cut the major F2 fold and have near-vertical orientations (Section 4j); and finally (v) outcrop of Drygill Shales (mid-Caradoc, Longvillian) within the igneous complex (Figure 12) is intruded by felsite and cut by the dykes (Eastwood *et al.*, 1968); this formation was deposited on the eroded F2 antiform (the shales now occur at a structural level well below the EVG) but was folded prior to dyke emplacement. These diverse lines of evidence indicate that both magmatism and deformation were concentrated in Early to Middle Caradoc times.

The Carrock Fell Granophyre and Round Knott Dolerite remanence directions clearly require adjustment for deformation of the post-Lower Carboniferous rocks. Since residual effects of Acadian deformation cannot be discerned, the palaeomagnetic results from these two bodies are not first-rank data. They do, however, have two important regional implications. (i) Since their *in-situ* magnetizations correspond most closely with the EVG, BVG and fully tilt-adjusted Carrock Fell Gabbro results, emplacement at a late stage in the Middle to Upper Ordovician cycle occurred after the bulk of regional deformation was complete. (ii) Both granophyre and dolerite record the first evidence for reversed polarity in the Ordovician magmatic activity; hence they place the major phases of Lake District Ordovician magmatism close to a polarity boundary following the exclusive normal polarity observed in the major phases of magmatic activity (EVG, BVG, Carrock Gabbros and Great Cockup Picrite, see Section 5a).

4h. The Great Cockup Picrite intrusion, northern Lake District

A suite of intrusions emplaced into the Skiddaw Slate north of Bassenthwaite in the northern Lake District (Figure 7) is linked by common feldspar and mafic compositions, and considered to be a part of a single consanguinous group (Eastwood *et al.*, 1968). Sample sites 1 and 2 (see Table 7 and Figure 7) are from separated outcrops in the same (Great Cockup) picrite intrusion (Postlethwaite, 1892) and site 6 is from the smaller Little Knott Picrite body. The remaining sites are in bodies of dioritic petrography although all have lamprophyric affinities (Eastwood *et al.*, 1968). Three K–Ar hornblende ages on the Great Cockup Picrite yield an average age of 458 ± 9 Ma (Rundle, 1979), with the oldest determination (468 ± 10 Ma) identical to the oldest K–Ar biotite age from the Carrock Fell Gabbro. The dioritic bodies have a geochemistry relating them to the Eycott Volcanic Group (Firman, 1978); thus the sampled intrusions 1 to 8 in Figure 7 are all linked to the Carrock–Borrowdale–Eycott igneous episode.

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Table 7. Site and group mean palaeomagnetic results, picrite and diorite intrusions, Bassenthwaite district, northern Lake District

Site no.	Grid ref.	N/n	D (*in-situ*)	I (*in-situ*)	D (Tilt-adjusted)	I (Tilt-adjusted)	R	k	α_{95}
1	3272 5324	17/10	45·6	−13·5	48·2	−48·3	9·01	9·1	17·0
2	3267 5328	17/13	46·5	13·1	46·8	−21·7	12·51	24·7	8·5
3	3223 5325	11/9	37·4	3·8	37·0	−31·2	8·51	16·2	13·2
Great Cockup Picrite (sites 1 and 2), group mean result									
31 samples			42·7	3·2	43·2	−31·8	28·56	12·3	7·7
(Palaeomagnetic Pole: 315·2°E, 9·3°S, d_p/d_m = 3·9/7·7°)									

The grid square is NY; other symbols are as for Table 2.

Petrographic examination shows all olivine in the picrite intrusions (sites 1, 2 and 6) to be altered to serpentine; Ti-augite is also variably altered. Primary compositional zoning and ophitic textures are, however, preserved and primary hornblende is not visibly altered; this latter mineral contains abundant opaque inclusions. More acid members including the diorites are more profoundly altered, especially the *c*. 100 m thick Embleton sill at site 8, where both hornblende and biotite are partially converted to chlorite; opaques are in large skeletal aggregates associated with alteration and likely to be secondary. All intrusions contain secondary quartz whilst the diorites are also extensively veined with carbonate.

These intrusions comprise outcrop near the edge of the Skiddaw Granite aureole (Figure 7) and the only discernible non-viscous components comprise overprints in a direction comparable to the batholith. The dominant feature of remanence in the Great Cockup Picrite, however, is a magnetite-resident component with NE declination and shallow inclination recovered following removal of a present field magnetization or a batholith overprint (samples 1-16 and 2-14 in Figure 17). All samples with this ChRM have the same polarity (Figure 17) and (excluding probable Skiddaw Granite overprints with $D > 70°$E, the A3 magnetizations in Figure 17) yield a group-mean direction of D/I = 42·7/3·2° (Table 7).

The only analogous palaeofield directions are of Upper Palaeozoic age but of opposite polarity (S-SW shallow) and were acquired during the Permo-Carboniferous reversed superchron; hence the magnetization in the picrite appears to predate deformation. The pluton is emplaced into folded Skiddaw Slates dipping NE

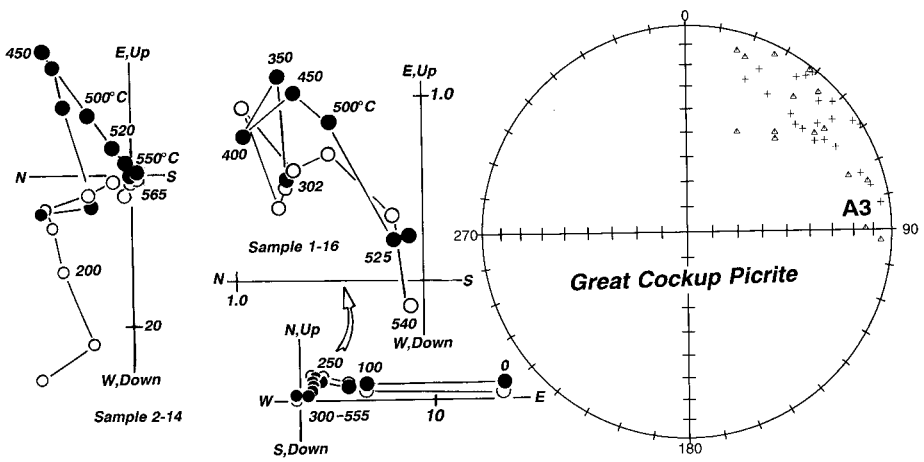


Figure 17. Typical results of progressive thermal demagnetization of samples from the Great Cockup Picrite intrusion (sites 1 and 2, see Figure 7 for locations) shown as orthogonal projections and using the same symbols as in Figure 4. The equal-area projection shows the distribution of sample ChRMs in the intrusion (*in situ*). Note that the only non-viscous component recognized in the other basic intrusion sites shown in Figure 7 was an A3 overprint of the regional batholith

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(Eastwood *et al*., 1968) and folded during F2 deformation; adjustment for tilt (35/40°E) of the regional fold (no appreciable plunge is evident on the limb of this F2 structure) moves the mean to $D/I = 43{\cdot}2/-31{\cdot}8°$ (31 samples, $\alpha_{95} = 7{\cdot}7°$). The remanence direction is therefore similar to the pre-F2 remanence in the EVG overlying the slate outcrop immediately to the north (Figure 7), although the declination is *c*. 30° more easterly. There is no apparent tectonic explanation for this difference because the same tilt adjustment is applicable to both the Great Cockup Picrite and the northern part of the EVG outcrop; hence it is interpreted to record polar movement between remanence acquisition of the picrite and the EVG (Section 5b).

4i. The Threlkeld–St John's Microgranite

This body is emplaced close to the contact between the Skiddaw Slates and BVG (Figure 1) and exposed as three small outcrops, two of which have been extensively quarried. The microgranite contains xenoliths of both Skiddaw Slate and Borrowdale Volcanics (Rastall, 1940), whilst the typical sub-volcanic nature of the pluton indicates a genetic link with late stages of Borrowdale activity. A Rb–Sr isochron age of 438 ± 6 Ma (Wadge *et al*., 1974; Rundle, 1981), equivalent to a late Ashgill stratigraphic age, also indicates that the microgranite was emplaced as a late phase of the Ordovician magmatism.

Thermal demagnetization of 116 individually oriented cores distributed over quarry outcrops of the microgranite and contacts isolates a component structure in weak magnetizations (*c*. 10^{-7} A m^2/kg) resident in magnetite. The ChRM component has positive inclination and southerly direction; with the exception of an l.b.t. component removed in the first step of treatment in some samples, it appears to be discrete (Figure 18a). Samples of Skiddaw Slate within a few metres of the margin of the pluton exhibit a component with the same direction. A few samples show weak evidence for reversal in the highest part of the spectrum (e.g. contact slate sample shown in Figure 18b, although only one acceptable component trajectory was resolved; no coherent components were isolated from slate samples further from the contact.

Sixty-seven samples possessed stable components of non-viscous origin. ChRM directions are predominantly SSW positive (Figure 18c and Table 8) and this population yields a mean direction of $D/I = 236{\cdot}5/63{\cdot}3°$ (41 samples including five in contact slates, $\alpha_{95} = 4{\cdot}7°$). Although this declination is typical of the Late Silurian–Early Devonian regional field direction for Britain, the inclination is nearly 30° steeper

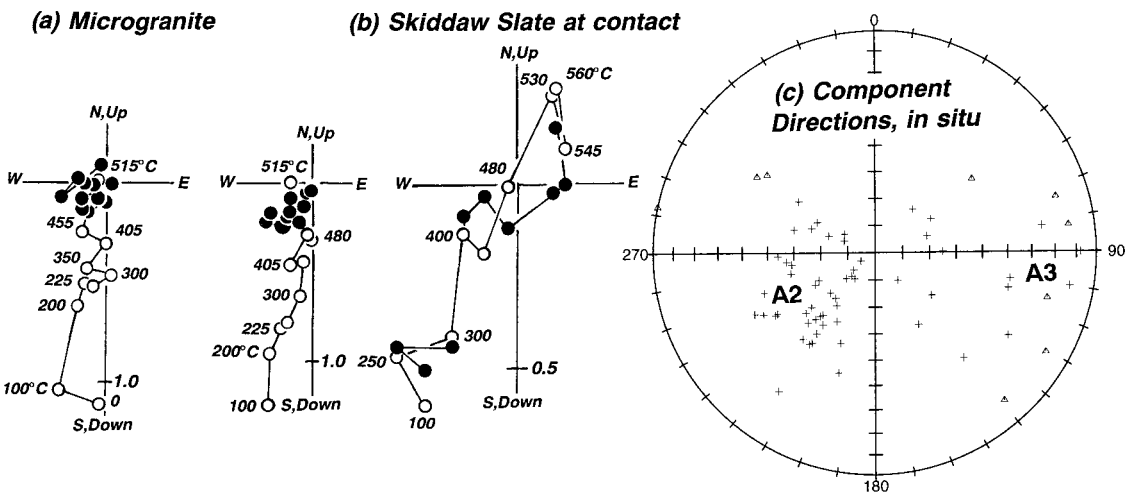


Figure 18. Results of progressive thermal demagnetization of samples from the Threlkeld–St John's Microgranite intrusion (*in situ*). (a) Orthogonal projections illustrating typical demagnetization of samples from the igneous body. (b) Demagnetization of a sample of Skiddaw Slate *c*. 3 m from the contact with the intrusion. (c) Equal-area plot of ChRM components in the pluton and contact slate (*in situ*). Symbols are as for Figure 4

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Table 8. Group mean palaeomagnetic results from the Threlkeld–St John's Microgranite and contact slate

	D (*in-situ*)	I (*in-situ*)	D (tilt-adjusted)	I (tilt-adjusted)	R	k	α_{95}
(i) SSW population							
41 samples	236·5	63·3	221·2	58·1	39·32	23·6	4·7
(Palaeomagnetic Pole: 317·6°E, 20·3°N, d_p/d_m = 5·9/7·4°)							
(ii) A3 (Devonian overprints)							
10 samples	97·5	11·2	99·1	8·1	9·08	9·8	16·2
(Palaeomagnetic Pole: 76·9°E, 1·9°S, d_p/d_m = 8·3/16·5°)							

Symbols are as for Table 2. The pole position is calculated on the assumption that this intrusion post-dates regional F2 folding (see text).

and, together with the evidence for overprinting at the contact with the slate country rock, suggests a primary remanence related to igneous cooling.

BVG lavas in the adjoining outcrop to the south have low southerly dips (*c*. 10°/170°E); hence tilt adjustment shifts the remanence direction only slightly to D/I = 221·2/58·1°. The Rb–Sr age implies that the pluton is *c*. 20–25 Ma younger than the Carrock Fell Gabbro; it is also predominantly of opposite (reversed) polarity thus contrasting with the predominant normal polarities of the Carrock Fell Complex, BVG and EVG. Since the Carrock Fell Complex was being emplaced during regional F2 folding (Section 4g), remanence in the Threlkeld–St John's body is likely to post-date Late Ordovician folding. The magnetization is then equivalent to a palaeolatitude of 45°S and a pole position at 318°E, 20°N; the declination is rotated by *c*. 40° anticlockwise from remanence in the Caradocian rocks and identifies terrane rotation between *c*. 465 and 440 Ma. Twelve sample components in the pluton with aberrant directions are of shallower inclination and most have easterly directions (Figure 18c); a correlation with A3 implies a patchy overprinting during regional batholith emplacement.

4j. Post-Carrock Fell Complex dyke swarms

Dykes of quartz dolerite and 'porphyrite' composition are emplaced into the Carrock Fell Complex and adjoining Skiddaw Slates and Eycott Volcanic Group (Eastwood *et al*., 1968). The spatial relationship to the complex and absence of high-level intrusion in this region during Silurian times identify them as a late phase of the Ordovician magmatic activity (no post-Ordovician volcanic activity is identified in this region with the exception of a thin suite of basaltic lavas at the base of the Carboniferous succession at the western margin of the outcrop in Figure 7 (Oppenheim *et al*., 1993)). Trends range from ESE–WNW to E–W (mostly within the Carrock Fell Complex) to ENE–WSW.

Four of five sampled dykes cutting the Eycott Volcanic Group at Eycott Hill (sites 1–4 in Figure 7 and see dyke sample in Figure 13) have NE–SW trends and yield discrete NE-directed ChRMs with negative inclination. A normal polarity field direction compatible with the Middle Devonian–Early Carboniferous palaeofield is therefore resolved from these members. The remaining dyke (5), with a SE trend more comparable to the Carrock dykes, has a contrasting remanence with similar declination but positive inclination (stereoplot in Figure 19 and Table 9i). A N to NE ChRM resident in magnetite is also observed in the majority of the dykes in the Carrock Fell region, including one (site 13) at the summit cutting the granophyre (Figure 19). When included with results from study of dykes in the same swarm to the northeast (Piper *et al*., 1978), three sites are observed to possess shallow inclination ChRMs suggesting that either (i) more than one episode of dyking is recorded here, or (ii) the E–W dykes have recorded non-dipolar field behaviour such as might have occurred if they were emplaced and magnetized during a reversal of the ancient field.

Since all dykes are substantially vertical, no major post-emplacement deformation is apparent at outcrop and they are adjusted in Table 9 only for the low tilts of Carboniferous strata above the adjoining

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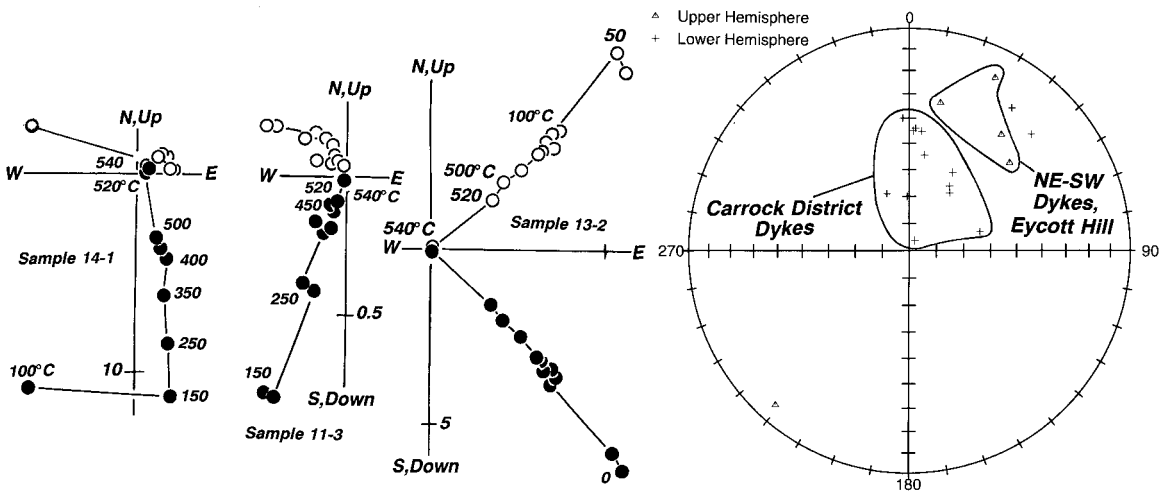


Figure 19. Orthogonal projections showing thermal demagnetization of samples from dolerite dykes cutting the Skiddaw Slates and Carrock Fell Complex; symbols are as for Figure 4. The equal-area projection shows site-mean directions of magnetization derived from individual dykes in this and a previous study (Piper *et al.*, 1978) and highlights the two swarms distinguished by their directions of magnetization

Table 9. Site and group mean palaeomagnetic results from dykes cutting the Lower Palaeozoic outcrop, eastern Lake District

Dyke no.	Grid ref.	N/n	D (*in-situ*)	I (*in-situ*)	D (tilt-adjusted)	I (tilt-adjusted)	R	k	α_{95}
(i) Eycott Hill									
1	3383 5301	8/8	15·6	−25·0	12·0	−32·5	7·90	81·4	6·2
2	3382 5293	7/7	40·0	−24·2	38·5	−33·8	6·95	116·1	5·6
3	3386 5292	8/5	49·5	−29·9	48·8	−39·9	4·80	19·9	17·6
4	3387 5292	8/7	27·3	−5·2	26·5	−14·0	6·81	31·3	10·9
5	3388 5291	9/9	34·1	31·5	35·8	22·1	8·87	62·6	6·6
(ii) Carrock–Blencathra									
9	3362 5306	8/5	220·2	−20·3	221·1	−8·7	4·90	39·9	12·3
10	3365 5342	6/6	352·1	50·5	3·1	44·0	5·73	18·0	16·2
11	3363 5345	6/5	323·5	73·3	359·1	70·0	4·69	12·7	22·3
12	3364 5346	3/2	257·0	77·9	338·6	87·5	1·91	11·5	—
13	3364 5355	12/7	45·4	36·9	46·5	25·1	6·83	34·9	10·4
14	3364 5355	6/3	248·2	81·7	29·2	85·7	2·89	18·7	29·4
15	3352 5350	10/10	348·8	43·9	357·1	40·0	9·97	356·1	2·6
Group mean results									
1. Quartz dolerite (ESE–WNW) dyke swarm									
(14 dykes)					21·4	56·6	12·95	12·4	11·7
(Palaeomagnetic Pole: 128·3°E, 67·2°N, d_p/d_m = 12·3/17·0°)									
2. Eycott dolerite (NE–SW) dykes									
(4 dykes)					30·9	−30·8	3·86	21·9	20·1
(Palaeomagnetic Pole: 326·5°E, 14·1°S, d_p/d_m = 12·5/22·4°)									

The group mean result from the quartz dolerite (ESE–WNW) swarm includes results from seven a.f. demagnetized dyke sites reported in Piper *et al.* (1978). Symbols used are as for Table 2.

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unconformity to the east. Remanence directions do not correlate with post-Acadian field directions from Britain, and the group-mean direction (Table 9) is equivalent to a palaeolatitude of 32°S lying between Ashgill and mid-Silurian determinations from this study (Sections 4c and 4i). This would therefore be compatible with an age of emplacement late in the Ordovician igneous episode; however, declinations are anomolous with respect to all the other Late Ordovician units and are not readily explained as a record of the geocentric axial dipole field.

5. ANALYSIS AND DISCUSSION

5a. Lower Palaeozoic apparent polar wander path of the Lake District terrane

Although the granitic rocks associate with batholith emplacement beneath the Lake District inlier are not ideal for establishing the palaeomagnetic imprint of this event late in the Caledonian Orogeny, a large number of samples from aureoles and the Shap Granite indicate that the regional geomagnetic field was a shallow dipolar axis directed approximately E–W. This can be compared with a NE−/SW+ field axis identified from Late Silurian–Early Devonian ('Old Red Sandstone') molasse adjoining the Caledonian orogen and from associated volcanics (Sallomy and Piper, 1973; Torsvik, 1985; Torsvik *et al*., 1991; Channell *et al*., 1992b; Setiabudidaya *et al*., 1994); in the Welsh Borderlands this palaeofield is observed to predate regional Acadian deformation (Channell *et al*., 1992b). The late Early Devonian–early Middle Devonian axis recorded in the Lake District terrane, which post-dates all but the late brittle stage of Acadian deformation, is rotated clockwise by *c*. 40° and is of shallower inclination; it implies that Britain had rotated further counterclockwise and moved into equatorial latitudes just south of the equator between Siluro-Devonian and late Early Devonian times (Table 10). The new data substantiate the westerly loop in the Siluro-Devonian APW path for Laurasia (Figure 20; Piper and Poppleton, 1990) but enhance its depth.

In the context of the likely volume and concomitant thermal anomaly associated with the regional batholith (Figure 1), the overprinting identified in the older Ordovician igneous rocks is remarkably small and recognized in isolated samples rather than as a ubiquitous partial thermal remanence (PTRM). PTRM acquisition is a function of both the amplitude and duration of the thermal anomaly (Pullaiah *et al*., 1975). Geological evidence from Shap shows that uplift and unroofing following the last Caledonian (Acadian) tectonic episode was relatively rapid and confined to Devonian times. However, the igneous activity was associated with fissure metasomatism and low-temperature hydrothermal mineralization along joints and fractures (e.g. Shap) and along the roof zones of the plutons (Skiddaw (see O'Brien *et al*., 1985)). Thus sporadic effects of late Early Devonian overprinting in the Ordovician rocks is probably attributable to permeation by these hot solutions.

The new palaeomagnetic results from the Carrock Fell Complex merit close comparison with results from the adjoining EVG. A positive fold test on components recognized in 27 lavas from stepwise a.f. cleaning and endpoint analysis (Morris, 1973; Briden and Morris, 1973) indicates that ChRMs in these rocks predate Late Ordovician F2 folding (Figure 7). Subsequent thermal demagnetization and component analysis has supported this conclusion (Piper *et al*., 1997). The ChRM in the EVG succession is therefore pre-Late Ordovician in age. Faunal evidence suggesting that emplacement of the EVG predated the BVG and occurred in Early Llanvirn times (Downie and Soper, 1972) has since been revised by Millward and Molyneux (1992), who show that no such tight constraint is provided by the geological evidence.

Directions of magnetization in the EVG are close to those in the lower part of the BVG and the case for correlating the two successions is reinforced by the common (normal) polarity found throughout the EVG and BVG (Piper *et al*., 1997). The polarity history of Llanvirn and Llandeilo times appears to comprise frequent reversals and a dominance of reversed polarity (Khramov and Rodionov, 1980; Farr *et al*., 1993); in contrast, the lower part of the Caradoc Stage was characterized by uniform normal polarity (Torsvik and Trench, 1992) embracing the *Nemagraptus gracilis* and early part of the *Diplograptus multidens* Biozones (late Llandeilo and early Caradoc). This is the single polarity found throughout 2500 m of EVG and

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Table 10. Ordovician–Devonian palaeomagnetic poles, Lake District and Welsh Basin terranes, British Paratectonic Caledonides

Code	Formation	Age	Pole °E/°N	d_p/d_m	D/I	Palaeolatitude	Quality criteria							Q	Reference
							1	2	3	4	5	6	7		
(i) Lake District Terrane															
GCP	Great Cockup Picrite	Om-u	315/ − 9	4/8	43/ − 32	17°S		x	x		x		x	4	This paper
EVG1	Binsey Volcanic Group‡	Om-u	347/ − 5	6/9	12/ − 49	30°S	x	x	x	x	x		x	6	Piper *et al.* 1997‡
EVG2	High Ireby Volcanic Grp‡	Om-u	359/ − 14	7/11	358/ − 38	21°S	x	x	x	x	x		x	6	Piper *et al.* 1997‡
BVG1	Borrowdale Volcanic Group	Ou, Cl	14/ − 6	4/6	341/ − 47	28°S	x	x	x	x	x		x	6	Piper *et al.* 1997
CFG1	Carrock Fell Gabbros	Ou, 469	344/5	3/5	17/ − 58	39°S	x	x	x	x	x		x	6	This paper
BVG2	Borrowdale Overprints	Ou	6/8	8/11	347/ − 62	43°S		x	x		x		x	4	Channell & McCabe 1992
CFG2	Carrock Fell Granophyre	(420)	24/20	11/13	319/ − 67	50°S		x	x		x	x	x	5	This paper
TSJ	Threlkeld–St John's Granite	438	318/20	6/7	237/63	45°S	x	x	x	x		x	x	6	This paper
CFG4	Carrock E–W Dyke Swarm§		308/ − 67	6/7	237/63	38°S		x	x		x		x	4	This paper§
CFG3	Round Knott Dolerite		339/ − 4	5/9	201/48	29°S			x		x	x	x	4	This paper
HF	Mudstones, Howgill Fells,	S, L1	310/ − 14	7/13	43/ − 24	14°S		x	x			x		3	Channell *et al.* 1993
BIE	Basic Intrusions, Ennerdale	429	2/ − 22	5/10	175/26	12°S		x	x	x	x	x		5	This paper
CFG5	NE–SW Dykes, Eycott		327/ − 14	13/22	31/ − 31	17°S	x	x	x		x			4	This paper
SH1	Skiddaw Hornfels	399	259/9	7/14	282/6	3°S	x		x		x	x	x	5	This paper
SG1	Shap Granite	393	264/7	2/5	278/17	9°S	x	x	x	x	x	x	x	7	This paper
(ii) Welsh Basin															
SVM	Stapeley Volcanic, Member*	Om (L1)	36/27	7/8	116/68	51°S		x	x		x		x	4	McCabe & Channell 1990
BV1	Builth Volcanics, south	Om (Lm)	4/ − 3	7/10	171/54	35°S	x	x	x	x			x	5	Trench *et al.* 1991
BV2	Builth Volcanics, south	Om (Lm)	4/ − 16	4/6	178/ − 43	25°S	x	x	x	x			x	5	McCabe *et al.* 1992
MYG	Moel-y-Golfa Andesite	Ou (Cm)	50/11	5/8	294/ − 50	31°S	x		x	x	x		x	5	Piper 1995
BV3	Builth Volcanics, north†	Om-u	−3/18	14/17	154/70	54°S	x	x	x	x	x		x	6	McCabe *et al.* 1992
SD	Shelve Dolerites	Ou	46/20	7/9	292/ − 59	58°S	x	x		x	x	x	x	6	Piper 1995
BDC	Breidden Dolerite Complex	Ou	22/27	19/22	314/ − 73	41°S	x	x	x		x	x	x	6	Piper 1995
TYG	Tan y Grisiau Granite	Ou	37/15	5/7	125/60	40°S	x	x	x		x	x	x	6	Piper *et al.* 1995

*Magnetization overprinted, probably by pervasive regional dolerites (c.f. SD and BDC).
†Result includes Late Ordovician dolerites comparable to SD and BDC.
‡Group mean calculation combining a.f. demagnetized sites (Briden and Morris, 1973) plus thermally demagnetized sites (Piper *et al.*, 1997).
§Group mean derived from sites of this study and a.f. demagnetized sites of Piper *et al.* (1978).
The quality factor (Q) is assessed from seven parameters (Van der Voo, 1993) comprising: 1, well determined magnetization age with no reason to believe that, it is significantly different from the rock age; 2, sufficient number of samples ($N > 25$) and high enough precision ($k > 10$, $\alpha_{95} < 16°$); 3, demagnetization undertaken; 4, positive fold tests; 5, sufficient structural control with no suspicion of local rotations; 6, presence of reversals; 7, lack of similarity with younger palaeopoles.

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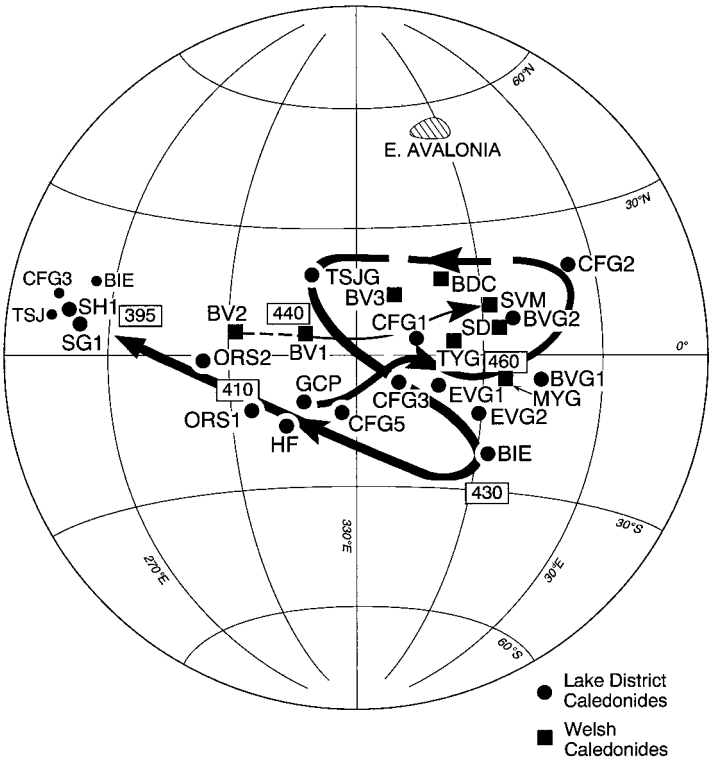


Figure 20. APW path for the Lake District terrane between mid-Ordovician and mid-Devonian times. The keys to the poles are given in Table 10. The Ordovician data from the Welsh Basin are omitted because they are rotated with respect to the Lake District data (see section 5.2 and Table 10). The codes used for the poles are given in Table 10; the small symbols are overprints of undefined age. Additional 'Old Red Sandstone' poles ORS1 and ORS2 are plotted to highlight early Devonian APW and come from the Welsh Basin (Channell *et al*., 1992b; Setiabudidaya *et al*., 1994)

more than 6000 m of the BVG (Piper *et al*., 1997). Hence palaeomagnetic evidence restores a traditional correlation between the two groups (Marr, 1900). The palaeomagnetic data imply that all major basic volcanism in the Lake District, including the intrusive activity at Carrock Fell, Great Cockup and elsewhere, belongs to a single Late Ordovician episode which followed final deposition of the Skiddaw Slates in Llanvirnian times but was completed by mid-Caradocian times. The normal polarity exhibited by these igneous events indicates that emplacement of all but the last activity at Carrock Fell (Carrock Granophyre, Round Knott Dolerite) occurred during the *Nemagraptus gracilis–Diplograptus multidens* normal-polarity chron.

Early results from the EVG were not adjusted for plunge of the regional F2 anticline; this omission was noted by Harris and Dagger (1987) and results are listed in Table 10 following this adjustment. They include new results (Piper *et al*., 1997) and are separated on the basis of data from the (lower) Binsey Formation and (higher) High Ireby Formation; the original study (Briden and Morris, 1973) noted a significant difference between results from these two formations.

The APW path for this terrane probably commences with the result from the Great Cockup Picrite; the pre-F2 deformation poles then define a counterclockwise rotation moving via the EVG poles to the BVG and Carrock Fell Layered Gabbro poles (Figure 20 and see also Figure 15). The latter result plots between primary and secondary magnetizations from the BVG; the collective data show an increase in palaeofield inclination during the earlier part of Caradoc times which is recognized in primary magnetizations from the BVG (Piper *et al*., 1997) and continues to the regional overprint (Channell and McCabe, 1992; Table 10). The path then

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continues with results from the Carrock Fell Granophyre (CFG2, also corresponding to migration into higher southerly latitudes, see Table 10), and the pole from the Threlkeld–St John's Microgranite with a radiometric age equivalent to the mid-Ashgill. Thus a 'hairpin' in the APW path correlates with the Late Ordovician deformation: poles on the anticlockwise segment are known, or probable, primary poles predating deformation whilst poles on the clockwise segment post-date deformation.

Continuing motion of this terrane into lower southerly paleolatitudes is recognized between Late Ashgill and mid-Silurian times (Table 10 and Figure 20). At this stage there seems to be a general accordance of results across Britain, embracing data from the 'Younger' granite suites (Torsvik, 1984; Piper, unpublished data) in the orthotectonic Caledonides in Scotland, to the Lake District (this work and Channell *et al.* 1993)), and to the border of the Variscan Belt in southern Britain (Torsvik *et al.*, 1993). Declinations are southerly and inclinations define shallow southerly paleolatitudes of *c.* 15°S. Terranes now comprising the British Isles are therefore assumed to have moved into close proximity with the Iapetus Ocean closed, but with final 'hard' welding of the Laurentian and Avalonian Plates still to be achieved during the Acadian Episode.

The subsequent APW path from the Lake District terrane moves via the mid-Silurian Howgill Fells pole to the late Early Devonian poles (Figure 20) and passes via the Late Silurian–Early Devonian poles noted above and resolved elsewhere in the British Caledonides; it highlights a westerly loop rapidly executed after mid-Silurian times. As noted elsewhere (Piper *et al.*, 1991; Trench and Torsvik, 1991; Channell *et al.*, 1992b), the pole had returned to a location close to mid-Silurian position by Middle or Late Devonian times. The westerly loop was therefore executed between *c.* 420 and 380 Ma and the hairpin corresponds approximately to the Acadian deformation.

5b. Comparison with the Welsh Basin: palaeogeography of the Caledonian margin of Avalonia

In Table 10 the summary of Ordovician–Devonian palaeomagnetic results from the Lake District terrane includes a quality factor (Q) assigned to each pole (Van der Voo, 1993). The Q factor cannot be a definitive indicator of the quality of a pole (the pole from the EVG, for example, is a result previously thought to be tightly constrained to a stratigraphic age), but it does give a good general indication of how satisfactory, or otherwise, a study is.

Ten Ordovician results with $Q \geqslant 3$ are now available from the Lake District terrane and seven results with $Q \geqslant 3$ cover the Middle–Upper Ordovician interval in the Welsh Basin. In each data set a location for Eastern Avalonia in intermediate southerly latitudes is recognized, with a motion into higher latitudes in later Caradocian times. However, there is significant contrast in declination (Table 10): whilst in both terranes the Caradocian declinations are approximately orthogonal to the Upper Ordovician fold axes, in Wales fold trends are between NNE–SSW and NE–SW whereas in the Lake District trends are approximately E–W. The mean difference between the declinations of primary Caradocian–early Ashgill magnetizations in the two regions is 55°. This difference evidently records later relative rotation between these two blocks (Figure 21). Since Acadian fold trends in both terranes are close to older Upper Ordovician trends, the rotation probably occurred after the culmination of the Acadian Episode as the Late Precambrian basement of Eastern Avalonia (the Midland Block (Thorpe *et al.*, 1984) and see Section 2 and Figure 21) impinged into the orogen prior to onset of horst and graben tectonics during Carboniferous times.

When the two terranes are restored to their Late Ordovician orientation (Figure 21), the Welsh Basin and Lake District volcanic arcs are seen to have comprised parallel lineaments at the time of their formation. Geological constraints which need to be incorporated into this reconstruction include the backarc origin of the Welsh province (Kokelaar, 1988) and the forearc origin of the Lake District province (Millward *et al.*, 1978). It is also likely that small terranes of greater antiquity such as the Padarn ridge and portions of Anglesey were sited in between. The combined arc assembly lay along the southern margin of the Iapetus Ocean; the palaeomagnetic data identify a latitudinal orientation in intermediate southerly latitudes during the major magmatic activity of Early Caradoc times (Figure 21).

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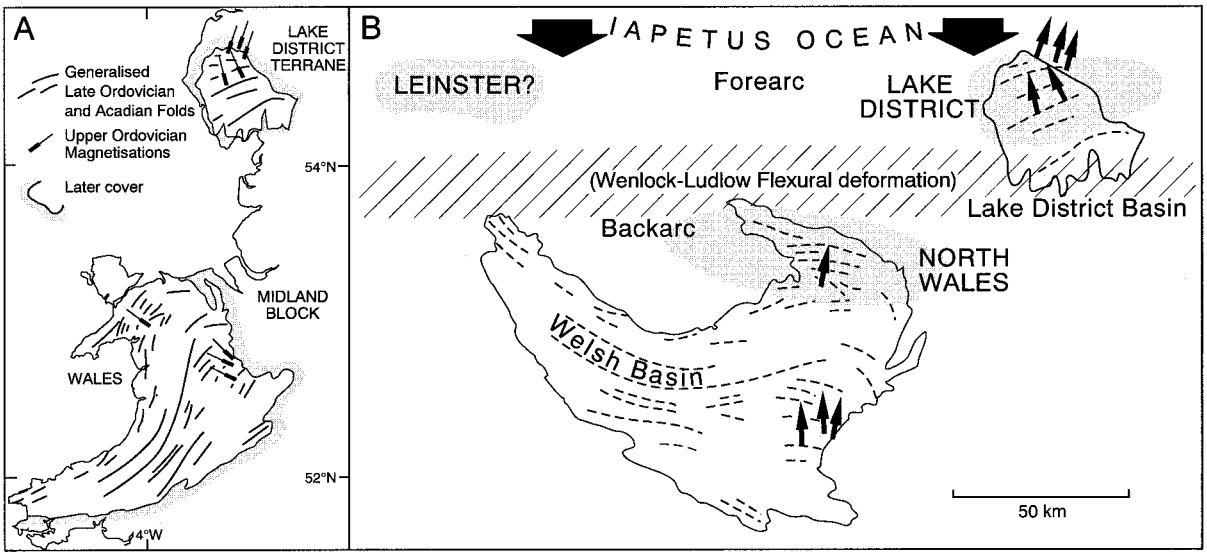


Figure 21. (A) Directions of Upper Ordovician (normal) magnetizations in the Welsh Basin and the Lake District terrane compared with the trends (generalized) of Upper Ordovician and Upper Silurian–Lower Devonian fold axes. The stippled border is the boundary of younger cover rocks. The magnetizations indicated are listed in Table 10 and are mostly of Caradocian age but may be as young as early Ashgill in some of the Welsh examples. (B) Palaeoreconstruction of the north British margin of Eastern Avalonia in Upper Ordovician and Silurian times adjusted for later rotation between the Welsh Basin and Lake District terrane and incorporating geological constraints noted in the text

Extinction of the volcanic arc during Ashgill times was followed by flexural deformation and development of narrow foreland basins as the Laurentian Plate overrode the smaller Avalonian Plate (Kneller, 1991). These unstable basins were infilled by turbidite deposition between Late Ordovician and Lower Devonian times. Owing to the paucity of suitable Silurian rocks for palaeomagnetic study, it cannot be positively confirmed that the configuration of Figure 21 was retained into the Silurian period. However, the Welsh Silurian Basin (now oriented NNE–SSW to NE–SW) is moved into parallelism with the Windermere Basin of the Lake District (now oriented approximately E–W) when adjusted for the 55° of relative rotation indicated by the Late Ordovician data; Acadian fold trends in the two basins are also brought into parallelism (Figure 22). The two basins are therefore inferred to have retained this configuration during basin infill and until the climax of Acadian deformation.

A lateral separation was also retained in Silurian times because infilling of the Lake District Basin took place predominantly by axial input from the east and commenced in Late Ordovician times (Kneller, 1991). Flexural deformation occurred later behind the arc in Wales because turbidite input commenced here during Llandovery times and major flexural subsidence was concentrated in late Pridoli and early Devonian times (King, 1994). Although axial input characterized both early and late stages of infill in both regions, input into the Welsh Basin was predominantly axial from the west (Figure 22). These input directions are recognized in both S (sole) structures and in magnetic fabrics of the turbidite units (Allinson, 1991; Piper *et al*., 1996). They are brought into alignment following adjustment for the 55° of rotation but highlight the opposing directions of dominant infill (Figure 22).

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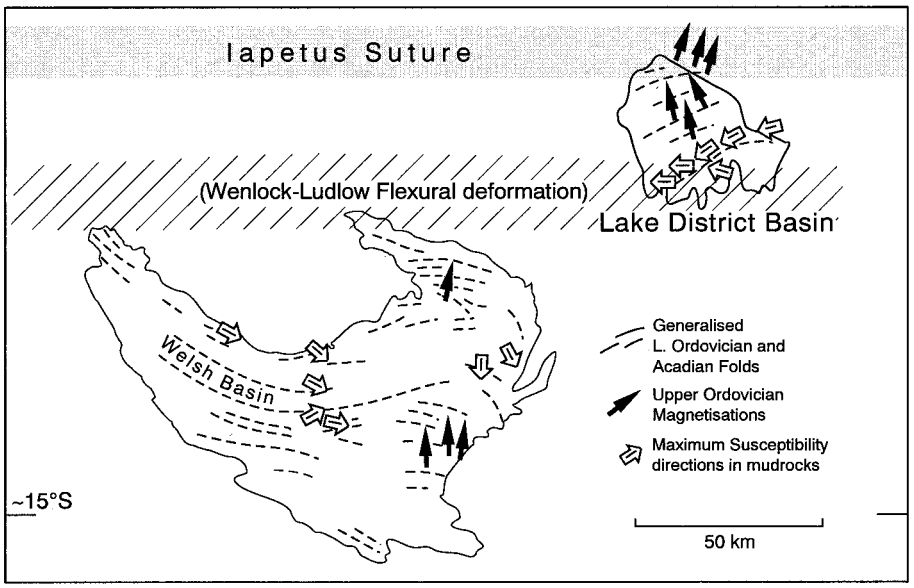


Figure 22. Palaeoreconstruction of the British Paratectonic Caledonides during Silurian times derived from Late Ordovician palaeomagnetic directions (black arrows) and geological constraints noted in the text. The mean directions of maximum susceptibility (k_1) axes shown by the open arrows are derived from turbidite units in the Silurian basins with directions of input as identified from nearby S structures; the magnetic fabric data are from Piper *et al*. (1996) for the Lake District Basin and from Allinson (1991) for the Welsh Basin

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